

Skytree GUI

Release 15.4.0

SKYTREE®

Notices

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Company Information:

Skytree, Inc.
1731 Technology Drive, Suite 700
San Jose, CA 95110
408.392.9300
www.skytree.net

Customers with a support contact can contact support using any of the following methods:

- Log in to support.skytree.net.
- Send questions to support@skytree.net.
- Call 408.392.9300 and select option 3.
- You can also view additional Skytree documentation by logging on to support.skytree.net.

Freemium customers can send inquiries to the LinkedIn® Community Group for Skytree:

<https://www.linkedin.com/groups/4468542>

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Chapter 1 Overview

Skytree is a machine learning platform that gives organizations the power to discover deep analytic insights, predict future trends, make recommendations, and reveal untapped markets and customers. As the only general purpose, scalable machine learning system on the market today, Skytree is built for the highest accuracy at unprecedented speed and scale. With a broad set of machine learning methods, Skytree works on massive, fast changing, and varied datasets in distributed computing environments with a high degree of efficiency and accuracy.

Skytree is comprised of two components: Skytree Server™ and Skytree Platform™. Skytree Server provides a CLI for performing data preparation and machine learning using state-of-the-art, fast, scalable modules. Skytree Platform includes a Python SDKs that provide access to REST API methods for performing classification, regression, and recommendation machine learning tasks. Skytree Platform also provides a graphical user interface for building models and viewing results.

About this Document

The Skytree GUI is a unified, project-based data scientist workspace for visually building, deploying, and managing models. The project-based approach allows users to easily manage data and activities to build models.

The Skytree GUI also improves model management and governance by providing self-documenting models and including a full audit trail. Models generated by the Skytree GUI automatically document all datasets, transformations, algorithms, and results, which are displayed in an easy to understand visual format. Additionally, all changes are tracked, creating a full audit trail, including the changes that were made when and by whom.

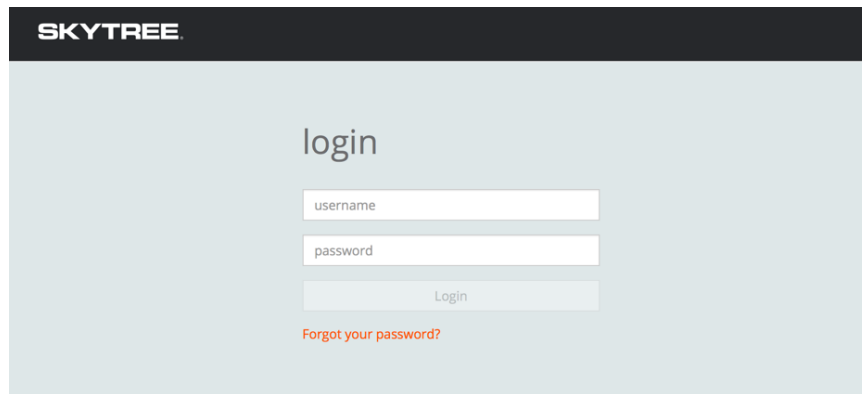
This document provides information for data scientists to Skytree to perform day-to-day tasks from the GUI including managing projects, uploading datasets, creating models using classification or regression methods, and viewing results.

Note: The Skytree GUI assumes that datasets are already in a Skytree-compatible format. Refer to the *Skytree Server User Guide*, the *Skytree REST API Guide*, or the *Skytree Python SDK Guide* documentation for information about data preparation.

Logging In and Logging Out

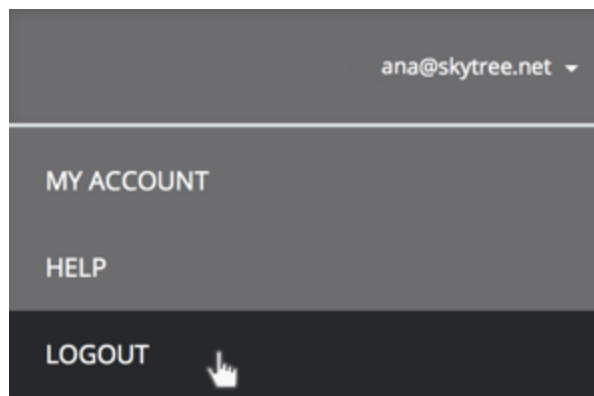
Logging In

After Skytree is installed, open a browser and navigate to the location where the Skytree GUI resides on your server. Enter the username and password that were provided to you by your Admin.

The image shows the Skytree login page. At the top is a dark header with the 'SKYTREE' logo in white. Below the header is a light blue-grey background. In the center, the word 'login' is displayed in a large, dark font. Underneath 'login' are three input fields: 'username', 'password', and a 'Login' button. Below the 'password' field is a link that says 'Forgot your password?' in a smaller, red font.

Logging Out

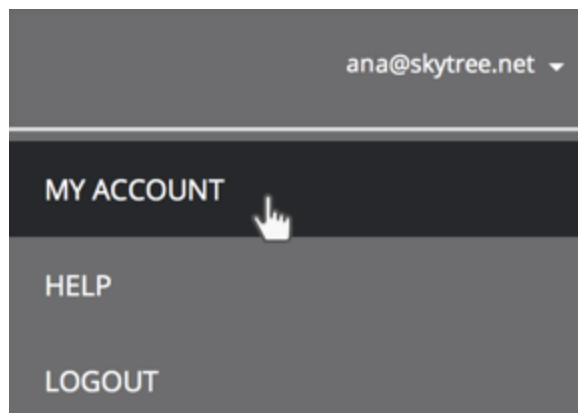
Click on the dropdown arrow beside your username in the upper-right corner, then select **LOGOUT** to end your session. In addition, the Skytree GUI will automatically log you out after 6 hours of inactivity.



Changing Your Password

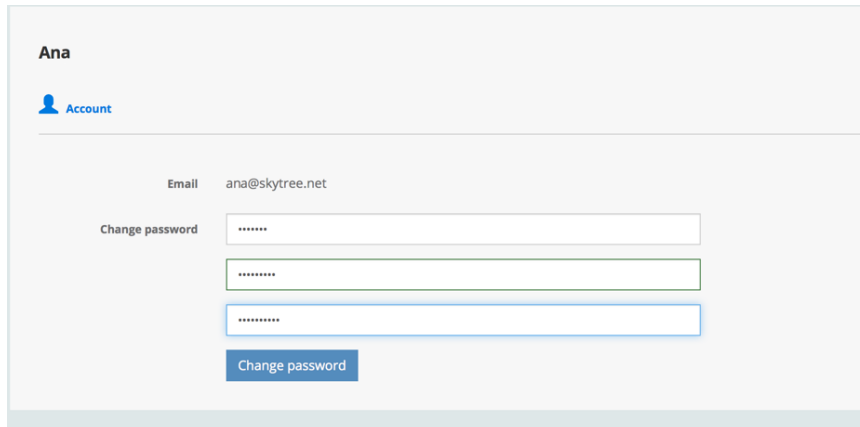
You can change your password from the My Account section of the Skytree GUI.

Click on the dropdown arrow beside your username in the upper-right corner, then select **MY ACCOUNT**.



This opens the My Account page, where you can specify a new password. Note that passwords must be a minimum of eight characters and contain at least one uppercase character, one lowercase character, and one number. Special characters are permitted but not required.

Note: Users who use LDAP/AD to authenticate cannot change their Skytree password.



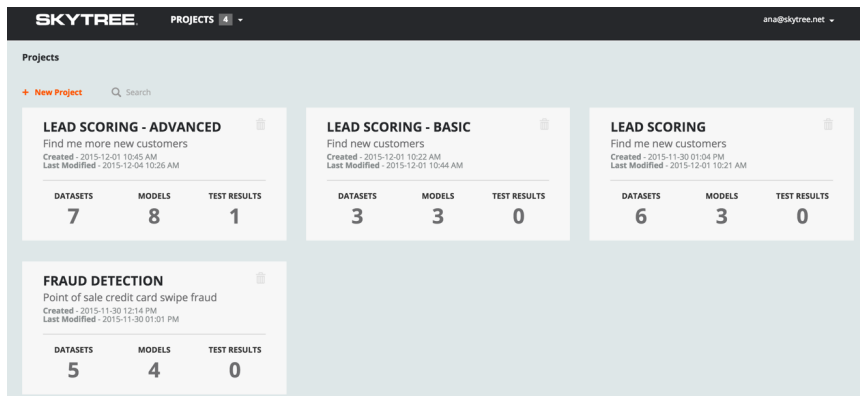
The screenshot shows the 'My Account' page for a user named 'Ana'. The page has a header with the user's name and a 'Account' link. Below this, the email address 'ana@skytree.net' is displayed. The 'Change password' section contains three password input fields (the first is highlighted with a blue border) and a 'Change password' button.

The Skytree GUI

The Skytree GUI opens to a Projects page that includes a card for each project (Project Cards). From this page, you can add a new project and search for projects. You can also click on a Project Card to view available information about datasets, models, and results that are included with the project.

For users who have a large number of projects, a **PROJECTS** drop down menu at the top of the page shows the three most recent projects.

If you know the name of a project, dataset, model, or test result that you want to view, you can type the name in the **Search** field, which is also available on the Datasets, Models, and Results pages. This search supports the wildcard expressions ***** and **?**.



The screenshot shows the Skytree GUI's 'Projects' page. The top navigation bar includes the 'SKYTREE' logo, a 'PROJECTS' dropdown menu, and the user's email 'ana@skytree.net'. Below the navigation bar, there's a 'Projects' section with a '+ New Project' button and a search bar. The main content area displays four project cards: 'LEAD SCORING - ADVANCED', 'LEAD SCORING - BASIC', 'LEAD SCORING', and 'FRAUD DETECTION'. Each card shows the project name, a brief description, creation and last modified dates, and a table of counts for Datasets, Models, and Test Results.

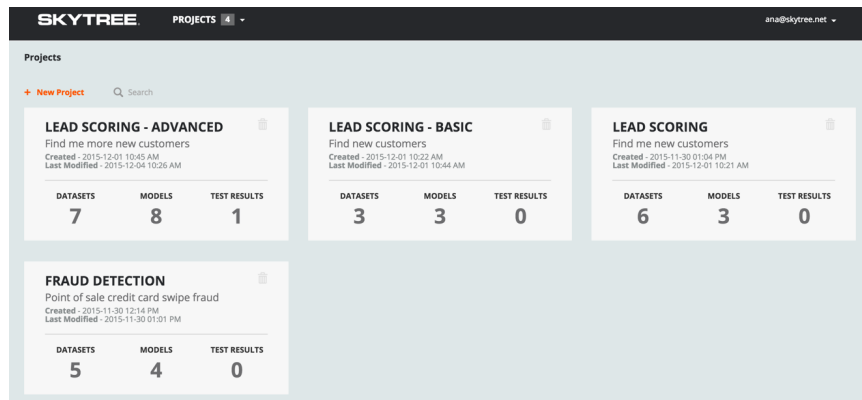
Project	Datasets	Models	Test Results
LEAD SCORING - ADVANCED	7	8	1
LEAD SCORING - BASIC	3	3	0
LEAD SCORING	6	3	0
FRAUD DETECTION	5	4	0

Chapter 2 Projects

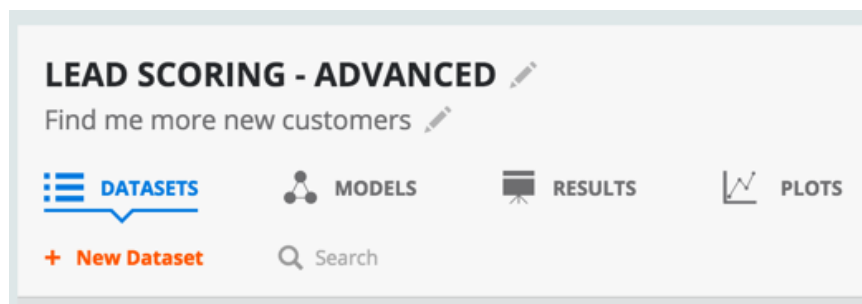
The Skytree GUI makes use of project-based machine learning. Whether you are trying to detect fraud or predict user retention, the datasets, models, and test results are stored and saved in the individual projects. The projects also retain a full audit trail for viewing how and when a dataset, model, or test result was added or built.

The Projects Page

The Projects page is the first that displays after successfully logging in to the Skytree GUI. This page shows Project Cards for each available projects; the number of available datasets, models, and test results in each project; and when each project was created/updated.



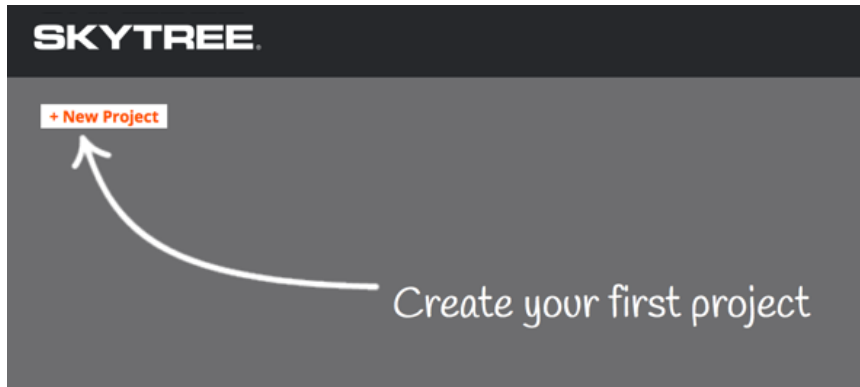
Click anywhere in a Project Card to view details of the included files. Click on Models to open the Models page and Test Results to open the Test Results page. Click anywhere else in a project card to view the Datasets page.



From this page, you can also edit the name and description of a project. Click the pencil icon (✎) beside the label that you want to change, and then enter a new name and/or description. Click the checkmark icon (✓) to complete the name change.

Adding Projects

The first time you log in to Skytree, your Projects page will be empty.



1. Click the **+ NEW PROJECT** button to start a new project.
2. Enter a unique name and a description for the project and then select **Create**. Note that you will receive an error if you attempt to specify a Project Name that already exists.

A screenshot of the 'New Project' form. The form has a light gray background and a title 'New Project' at the top left. It contains two input fields: 'Project Name' with a placeholder text 'Please provide a project name' and 'Description'. At the bottom right of the form, there are two buttons: 'Cancel' (gray) and 'Create' (blue).

Upon successful completion, the new Project Card will appear on the Projects page. At this point, you can begin adding datasets and then training/tuning/testing models.

Deleting Projects

On the Projects page, you can delete a project by clicking the delete icon (🗑) in the upper-right corner of a project card.

LEAD SCORING - ADVANCED

Find me more new customers

Created - 2015-12-01 10:45 AM
Last Modified - 2015-12-04 10:40 AM

DATASETS

8

MODELS

8

TEST RESULTS

1

FRAUD DETECTION

Point of sale credit card swipe fraud

Created - 2015-11-30 12:14 PM
Last Modified - 2015-11-30 01:01 PM

DATASETS

5

MODELS

4

TEST RESULTS

0

LEAD SCORING - BASIC

Find new customers

Created - 2015-12-01 10:22 AM
Last Modified - 2015-12-01 10:44 AM

DATASETS

3

MODELS

3

TEST RESULTS

0

After clicking the delete icon, you will be prompted to continue with the deletion.

Note: You cannot delete a project that includes any INPROGRESS items.

Delete Project?

Do you want to delete **Lead Scoring - Advanced**?

The following items will also be deleted:

- 8 dataset(s)
- 8 model(s)
- 1 result(s)

Delete

Cancel

Click **Delete** to delete the project. All data associated with the project will also be deleted.

Note: You can also delete a project from any page within the GUI by selecting the delete icon on the upper-right corner of the page that you are viewing.

Chapter 2 Projects

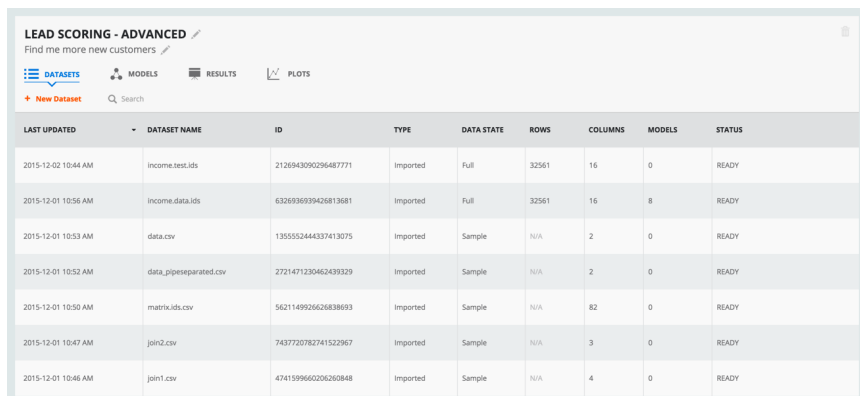
Skytree GUI | 7

Chapter 3 Datasets

Datasets represent structured source files. A dataset can contain only real columns, only categorical columns, or both real and categorical columns. If both types of columns are present, then the real columns must precede the categorical columns, i.e., the header reads “real:n,int:m”. Optionally, the dataset can consist of 3 meta columns representing the label column, the target column, and the ID column respectively.

Note: The Skytree GUI assumes that datasets are already in a Skytree-compatible format. Refer to the *Skytree Server User Guide*, the *Skytree REST API Guide*, or the *Skytree Python SDK Guide* for information about data preparation.

Click anywhere in a Project Card to view the available datasets and/or add a new dataset.



The screenshot shows the 'LEAD SCORING - ADVANCED' interface with a navigation bar containing 'DATASETS', 'MODELS', 'RESULTS', and 'PLOTS'. Below the navigation bar is a '+ New Dataset' button and a search bar. The main content is a table with the following columns: LAST UPDATED, DATASET NAME, ID, TYPE, DATA STATE, ROWS, COLUMNS, MODELS, and STATUS. The table contains 8 rows of dataset information.

LAST UPDATED	DATASET NAME	ID	TYPE	DATA STATE	ROWS	COLUMNS	MODELS	STATUS
2015-12-02 10:44 AM	income.test.ids	2126943090296487771	Imported	Full	32561	16	0	READY
2015-12-01 10:56 AM	income.data.ids	6326936939426813681	Imported	Full	32561	16	8	READY
2015-12-01 10:53 AM	data.csv	1355552444337413075	Imported	Sample	N/A	2	0	READY
2015-12-01 10:52 AM	data_pipe-separated.csv	272147123046249329	Imported	Sample	N/A	2	0	READY
2015-12-01 10:50 AM	matrix.ids.csv	5621149926626838693	Imported	Sample	N/A	82	0	READY
2015-12-01 10:47 AM	join2.csv	7437720782741522967	Imported	Sample	N/A	3	0	READY
2015-12-01 10:46 AM	join1.csv	4741599660205260948	Imported	Sample	N/A	4	0	READY

Adding Datasets

Click the **+ NEW DATASET** button to add a new dataset.

Projects > Prediction > Datasets > **New Dataset**

Location Register HDFS ▾

Name Register HDFS ✓
Import Local File

HDFS

☐ First Line Has Column Names

Separator Comma ▾

☐ Has Missing Value Identifier

[Preview](#)

Datasets can be added by registering an HDFS file on a cluster or by importing a local file.

Register an HDFS File

Perform the following to create a dataset by registering an HDFS file:

1. Select Register HDFS from the drop down menu.
2. Specify the name of the dataset.
3. Enter the HDFS URL.
4. Specify whether the first line includes column names (headers).
5. Select the data separator. Options include Comma, Semicolon, Tab, Space, or a User Defined separator (such as a "VBAR").
6. Specify whether the data in the file includes missing values. If it does, then specify the character in the dataset that indicates the missing values. This defaults to "?"

[illegible]

- After the data file is selected, a Raw Data View displays, showing the first several lines of data. (Note that this raw view is limited to 100,000 characters, including line breaks.) Click **Preview**, then scroll down to view the first 100 rows of the dataset in a tabular format. From this preview, you can optionally specify a column in this dataset that includes free text using the Text Blob Column drop-down. You can also specify the column in this dataset that will serve as the ID column when performing predictions and evaluations. For example, in the following image, `feature_2` is selected as the ID column, or the column to predict during Predict & Evaluate operations.

Note: An ID column is required when performing Predict & Evaluate operations. Text Blob columns cannot be used as ID columns.

First Line Has Column Names

Separator: Comma

Has Missing Value Identifier

Preview Import

Dataset Preview (showing rows 1-100)

Text Blob Column: No Selection

ID Column: feature_2

FEATURE_0	FEATURE_1	FEATURE_2	FEATURE_3	FEATURE_4	FEATURE_5	FEATURE_6
0	0	0.7071067811865475	0	0	0	0
0	0	0.7071067811865475	0	0	0	0
0	0	0.7071067811865475	0	0	0	0
0	0	0.7071067811865475	0	0	0	0
0	0	0.7071067811865475	0	0	0	0

- Click **Import** to add that dataset to the project. The Datasets page will show the status of the import in the right column. This status will initially show INPROGRESS. While in this state, you can click the **View Job Status** link to view additional details about the status of the dataset. The status will change to READY when the import completes successfully.

MODELS	STATUS
0	INPROGRESS : Sample . View Job Status
0	READY

Note: The **View Job Status** link will not work "out of the box" for Amazon's EMR users. These users will need to configure a SOCKS proxy to view the status link. Consult the [AWS documentation](#) for more information.

Import a Local File

Perform the following steps to import a local file.

- Click the **Select file** button.

2. Browse to the file location, then click **Choose**. Once selected, a preview of the raw data appears on the page. This preview is helpful in determining whether the dataset has column headers and determining the data separator and missing value identifier.
3. The Name field defaults to the file name. If desired, you can edit this file name.
4. Specify whether the first line of the file includes column names (headers).
5. Select the data separator. Options include Comma, Semicolon, Tab, Space, or a User Defined separator (such as a "VBAR").
6. Specify whether the data in the file includes missing values. If it does, then specify the character in the dataset that indicates the missing values. This defaults to "?"

Location: Import Local File ▼

File: Clear Selection

Raw Data View

```
age, workclass, fnlwgt, education, education-num, marital-status, occupation,
39, State-gov, 77516, Bachelors, 13, Never-married, Adm-clerical, Not-in-famil
50, Self-emp-not-inc, 83311, Bachelors, 13, Married-civ-spouse, Exec-manageria
38, Private, 215646, HS-grad, 9, Divorced, Handlers-cleaners, Not-in-family, W
53, Private, 234721, 11th, 7, Married-civ-spouse, Handlers-cleaners, Husband,
28, Private, 338409, Bachelors, 13, Married-civ-spouse, Prof-specialty, Wife,
37, Private, 284582, Masters, 14, Married-civ-spouse, Exec-managerial, Wife, W
```

Name: income.data

☒ First Line Has Column Names

Separator: Comma ▼

☒ Has Missing Value Identifier

Missing Value Identifier: ?

Preview

7. Once the data file is selected, a Raw Data View displays, showing the first several lines of data. (Note that this raw view is limited to 100,000 characters, including line breaks.) Click **Preview**, then scroll down to view the first 100 rows of the dataset in a tabular format. From this preview, you can optionally specify a column in this dataset that includes free text using the Text Blob Column drop-down. You can also specify the column in this dataset that will serve as the ID column when performing predictions and evaluations. For example, in the following image, the yearly-income column is selected as the ID column, or the column to predict during Predict & Evaluate operations.

Note: An ID column is required when performing Predict & Evaluate operations. Text Blob columns cannot be used as ID columns.

Name:

☒ First Line Has Column Names

Separator:

☒ Has Missing Value Identifier

Missing Value Identifier:

Dataset Preview (showing rows 1-100)

Text Blob Column: ID Column:

AGE	WORKCLASS	FNLGWT	EDUCATION	EDUCATION-NUM	MARITAL-STATUS	OCCUPATION
39	State-gov	77516	Bachelors	13	Never-married	Adm-clerical
50	Self-emp-not-inc	83311	Bachelors	13	Married-civ-spouse	Exec-manageria
38	Private	215646	HS-grad	9	Divorced	Handlers-cleanse
53	Private	234721	11th	7	Married-civ-spouse	Handlers-cleanse
28	Private	338409	Bachelors	13	Married-civ-spouse	Prof-specialty

- Click **Import** to add that dataset to the project. The Datasets page will show the status of the import in the right column. This status will initially show INPROGRESS. While in this state, you can click the **View Job Status** link to view additional details about the status of the import. The status will change to READY when the import completes successfully.

MODELS	STATUS
0	 INPROGRESS : Sample . View Job Status
0	READY

Note: The **View Job Status** link will not work "out of the box" for Amazon's EMR users. These users will need to configure a SOCKS proxy to view the status link. Consult the [AWS documentation](#) for more information.

Dataset List

Datasets that are successfully added to a project will display in a dataset table. Click on a column heading to sort the data in that column. Mouse over an entry in the list to view the following additional options:

- View details:** View additional information about the dataset, including raw file information.
- View logs:** Review the log files that were created when the dataset was added.
- Create model:** Create a new model based on this dataset.
- View graph:** View a graphical representation of how datasets in the project originated. If this dataset was used to created models and predictions, then those will be represented as well.
- Commit dataset:** Updates the dataset with recomputed metadata across the whole dataset rather than providing only a sample of the dataset. Note that this menu option will only appear for datasets that have not been committed.

- **Predict & Evaluate:** Using a dataset from this list, compare the dataset against an available model to predict and evaluate an ID column. Refer to [Adding Models](#) (page 23) for information on training models.
- **Transform:** Allows you to transform a dataset.
- **Delete:** Deletes the selected dataset.

LEAD SCORING - ADVANCED
Find me more new customers

[DATASETS](#) [MODELS](#) [RESULTS](#) [PLOTS](#)

[+ New Dataset](#)

LAST UPDATED	DATASET NAME	ID	TYPE	DATA STATE	ROWS	COLUMNS	MODELS	STATUS
2015-12-04 10:40 AM	horizontalized.data	32245	Imported	Sample	N/A	3	0	READY
2015-12-02 10:44 AM	income.test.ids	32771	Imported	Full	32561	16	0	READY
2015-12-01 10:56 AM	income.data.ids	33681	Imported	Full	32561	16	8	READY
2015-12-01 10:53 AM	data.csv	33075	Imported	Sample	N/A	2	0	READY
2015-12-01 10:52 AM	data_pipe_separate	33929	Imported	Sample	N/A	2	0	READY

Most of these options are described in the sections that follow. Refer to [Predict & Evaluate](#) (page 36) in the Results chapter for information on how to perform a Predict and Evaluate operation.

View Dataset Details

Select an entry from the list of datasets, then click **View details** to view additional information about the dataset.

From this page, you can select one of the following views for more detailed information:

- **Data:** View the raw data included in the dataset. In this view, header values from source become column names. This is the default view. Click on a column heading to sort the data in that column.

income.data.ids

Id: 3569680916978313175
Last Updated: 2015-09-14 09:46 AM
Rows: 32561
Data State: Full
Status: READY

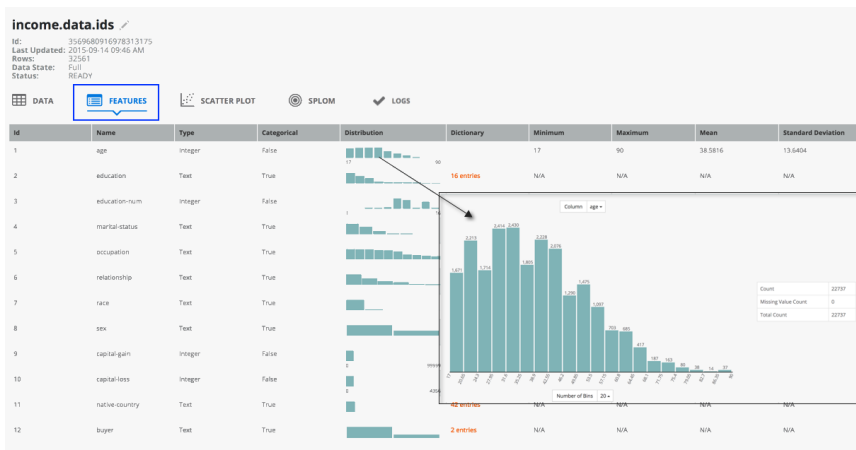
[DATA](#) [FEATURES](#) [SCATTER PLOT](#) [SPLIM](#) [LOGS](#)

age	workclass	fnlwgt	education	education-num	marital-status	occupation
39	State-gov	77516	Bachelors	13	Never-married	Adm-clerical
50	Self-emp-not-inc	83311	Bachelors	13	Married-civ-spouse	Exec-managerial
38	Private	215646	HS-grad	9	Divorced	Handlers-cleaner
53	Private	234721	11th	7	Married-civ-spouse	Handlers-cleaner
28	Private	338409	Bachelors	13	Married-civ-spouse	Prof-specialty
37	Private	284582	Masters	14	Married-civ-spouse	Exec-managerial
49	Private	160187	9th	5	Married-spouse-absent	Other-service

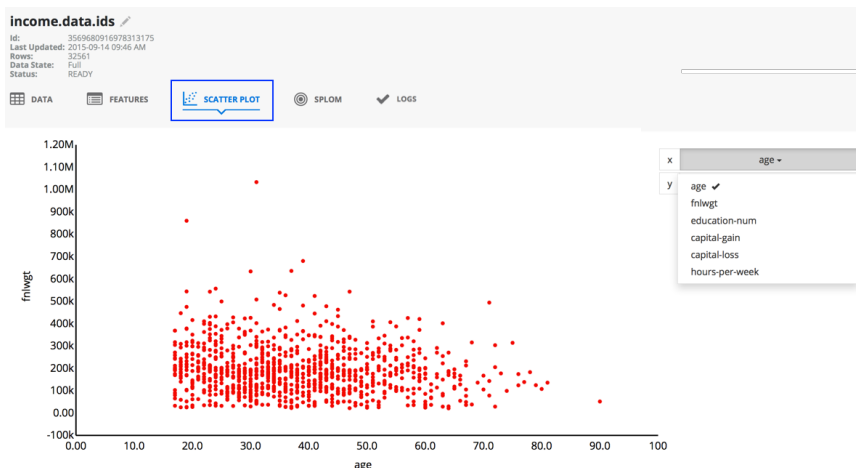
Note: If a text dataset was transformed, and a Text Blob column was specified, the data will display with "..." on the Dataset Details screen. For example:



- Features:** View the features of the dataset that will be included during model creation including min, max, mean, and sigma values, and whether the entry includes categorical data. For categorical data, this also shows dictionary entries. You can click on a distribution to view a larger histogram/barchart for each column in the original data. Within the histogram, you can select a different column, and you can select a different number of bins for numeric data.

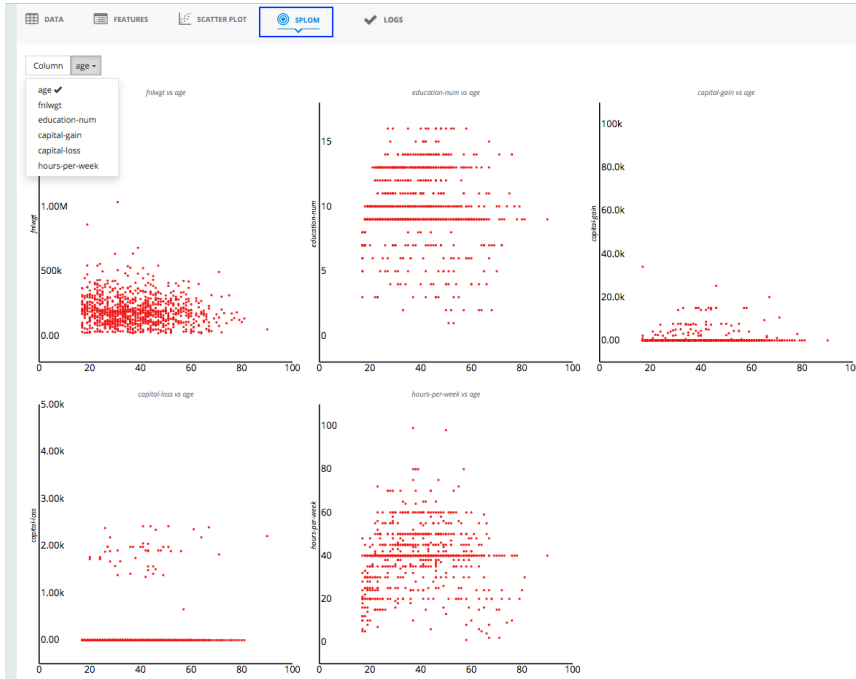


- Scatter Plot:** View an interactive graph that plots the dataset features. The data shown is a sampled subset of the data (top 1000 rows). You can change the view of the graph by specifying different X and Y axis features.



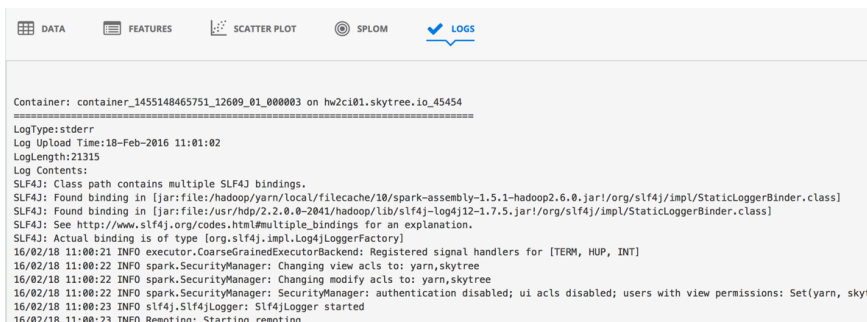
Note: The Scatter Plot might not show any points if the dataset includes only one or two numerical columns.

- **SPLOM:** View interactive SPLOM (scatter plot matrix) values. The data shown is a sampled subset of the data (top 1000 rows). You can change the scatter points that display by specifying a different target column for the X axis.



Note: The SPLOM graphs might not show any points if the dataset includes only one or two numerical columns.

- **Logs:** View logging information that was generated when the dataset was added to the UI. A log is available when the dataset is no longer INPROGRESS.



```
Container: container_1455148465751_12609_01_000003 on hw2c101.skytree.io_45454
LogType:stderr
Log Upload Time:18-Feb-2016 11:01:02
LogLength:21315
Log Contents:
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/hadoop/yarn/local/warehouse/10/spark-assembly-1.5.1-hadoop2.6.0.jar!/org.slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/hdp/2.2.0-2041/hadoop/lib/slf4j-log4j12-1.7.5.jar!/org.slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
16/02/18 11:00:21 INFO executor.CoarseGrainedExecutorBackend: Registered signal handlers for [TERM, HUP, INT]
16/02/18 11:00:22 INFO spark.SecurityManager: Changing view acls to: yarn,skytree
16/02/18 11:00:22 INFO spark.SecurityManager: Changing modify acls to: yarn,skytree
16/02/18 11:00:22 INFO spark.SecurityManager: SecurityManager: authentication disabled; ui acls disabled; users with view permissions: Set(yarn, skyt
16/02/18 11:00:23 INFO slf4j.Slf4jLogger: Slf4jLogger started
16/02/18 11:00:23 INFO Reentrant: Starting reentrant
```

From this page, you can also edit the name of the dataset. Click the pencil icon (✎) beside the name, and then enter a new name for the dataset. Click the checkmark icon (✓) to complete the name change.

✓ ✕

Id: 3569680916978313175
Last Updated: 2015-09-14 09:46 AM
Rows: 32561
Data State: Full
Status: READY

DATA
FEATURES
SCATTER PLOT
SPLM

View Logs

Select a dataset from the list, then click the **View logs** button to view logging information that was generated when the dataset was added to the UI. A log is available when the dataset is no longer INPROGRESS.

Click the links above the log to navigate to the Dataset Data, Features, Scatter Plot, or SPLM pages.

Note: You can also access the dataset log files when viewing dataset details by selecting the **Logs** link.

DATA
FEATURES
SCATTER PLOT
SPLM
✓ LOGS

Container: container_1455148465751_12609_01_000003 on hw2ci01.skytree.io_45454

LogType: stderr

Log Upload Time: 18-Feb-2016 11:01:02

LogLength: 21315

Log Contents:

```

SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/hadoop/yarn/local/warehouse/10/spark-assembly-1.5.1-hadoop2.6.0.jar!/org.slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/usr/hdp/2.2.0-2041/hadoop/lib/slf4j-log4j12-1.7.5.jar!/org.slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.slf4j.impl.Log4jLoggerFactory]
16/02/18 11:00:21 INFO executor.CoarseGrainedExecutorBackend: Registered signal handlers for [TERM, HUP, INT]
16/02/18 11:00:22 INFO spark.SecurityManager: Changing view acls to: yarn,skytree
16/02/18 11:00:22 INFO spark.SecurityManager: Changing modify acls to: yarn,skytree
16/02/18 11:00:22 INFO spark.SecurityManager: SecurityManager: authentication disabled; ui acls disabled; users with view permissions: Set(yarn, skyt
16/02/18 11:00:23 INFO slf4j.Slf4jLogger: Slf4jLogger started
16/02/18 11:00:23 INFO Remoting: Starting remoting

```

Create Model

Select a dataset from the dataset list, then click **Create model** to create a new model based on the selected dataset. This page is similar to the [Adding Models](#) (page 23) page, but the Dataset field is already pre-populated with the selected dataset name.

Dataset
Select Dataset

Model Name

Refer to [Adding Models](#) (page 23) for more information on how to create a model from a dataset.

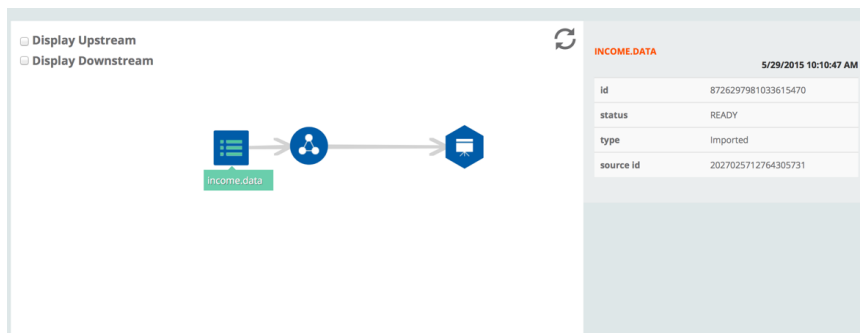
View Graph

Select a dataset from the dataset list, then click **View graph** to view graphical information about the dataset. If the dataset was used to transform additional datasets, then each of those transforms will also appear on the graph. If the dataset was used to build models and then used when generating a test result, then those icons will display as well. This provides a complete lifecycle of the dataset from import to result.

A dataset lifecycle can include the following information.

	This represents a dataset.
	This represents a model built from a dataset.
	This represents a Predict & Evaluate result.

The graphical information is interactive. When you select an icon in the graph, additional detailed information about that step displays in the right column. In the image below, you can see from the right column that the `income.data` file was imported. This file name is also a hyperlink. Click the file name to navigate to the dataset details. Use the **Display Upstream** and **Display Downstream** options to view the specific files that are related to this dataset.



The screenshot shows a graphical interface with a flow diagram on the left and a details panel on the right. The flow diagram shows a sequence of three icons: a dataset icon, a model icon, and a Predict & Evaluate icon, connected by arrows. The first icon is labeled 'income.data'. The details panel on the right is titled 'INCOME.DATA' and shows the following information:

5/29/2015 10:10:47 AM	
id	8726297981033615470
status	READY
type	Imported
source id	2027025712764305731

Commit a Dataset

When a dataset is initially added to a project, the dataset includes a sample of the entire data (as indicated in the DATA STATE column).

LAST UPDATED	DATASET NAME	ID	TYPE	DATA STATE	ROWS
5/15/2015 10:22:36 AM	Income.data	252788723262168710	Imported	Sample	N/A

You can specify to recompute the metadata across the whole dataset by committing the dataset.

1. Select a dataset from the list of datasets, then click **Commit dataset**. Note that this option only appears on datasets that have not yet been committed.
2. A confirmation box displays. Select **Commit** to continue the process or **Cancel** to end the process.

Commit Dataset

Dataset ID 252788723262168710

Dataset Name income.data

Upon successful completion, the DATA STATE will change to Full, and the ROWS column will populate with the appropriate information.

LAST UPDATED	DATASET NAME	ID	TYPE	DATA STATE	ROWS
5/15/2015 11:20:00 AM	income.data	252788723262168710	Imported	Full	32561

Transform Dataset

Skytree allows you to apply transformations to the datasets in your project directly through the UI.

Filter Columns:

NAME	UNIQUE ID	CATEGORICAL	BLOB	REMOVE
age	☐	☐	☐	☐
workclass	☐	☒	☐	☐
fnlwgt	☐	☐	☐	☐
education	☐	☒	☐	☐
education-num	☐	☐	☐	☐
marital-status	☐	☒	☐	☐

Preview

No changes

New Dataset Name:

Through the Skytree UI, you can perform the following transformations:

- Datasets can be filtered by rows to include a subset of data. The filter is based on a given Lambda expression, which can be added in the Filter Columns entry field.
- An ID column can be added using the **Generate ID Column** button. When selected, a new column appears at the top of the dataset table. You will be prompted to provide a name for this new column. This new column will include unique IDs.

Note: The **Generate ID Column** button is disabled if the dataset already includes an ID column.

- An existing column can be specified as a unique ID column. (Note that a dataset can only include one unique ID column.)
- Columns can be specified as categorical.

Note: For columns with more than 2048 categorical values, the column is converted to non-categorical text in order to limit the dictionary size.

- Non-categorical columns can be specified as blob columns.
- Columns can be removed.

When a transformation is selected, the UI provides the request information in the Preview box. For example, the following image specifies to add an ID column and to remove the fnlwtg column.

You can also specify a new name for the transformed dataset. Click **Transform Dataset** when you are finished.

NAME	UNIQUE ID	CATEGORICAL	BLOB	REMOVE
ID	☒	☐	☐	☐
age	☐	☐	☐	☐
workclass	☐	☒	☐	☐
fnlwtg	☐	☐	☐	☒
education	☐	☒	☐	☐
education-num	☐	☐	☐	☐

```

{
  "transform": {
    "inputs": [
      "1896550715375931489"
    ],
    "transformSteps": [
      {
        "columnName": "ID",
        "type": "AddUniqueIdColumn"
      },
      {
        "type": "RemoveColumns",
        "removeColumns": [
          "fnlwtg"
        ]
      }
    ]
  }
}

```

New Dataset Name:

Note: You can also transform a dataset from the Graph view by right-clicking on the dataset and selecting **Transform**.

Delete Datasets

Select an entry from the list of datasets, then click **Delete** to delete the dataset along with all downstream files that were generated from this dataset (for example, models and results).

Note: You cannot delete application type datasets (for example, datasets that were built from test results).

When you click **Delete**, Skytree will prompt you with to confirm. This confirmation box will include list of all files that the dataset was referenced in. Click **Delete** again to delete the dataset and all associated files.

Delete Dataset? ×

Do you want to delete **income.data**?

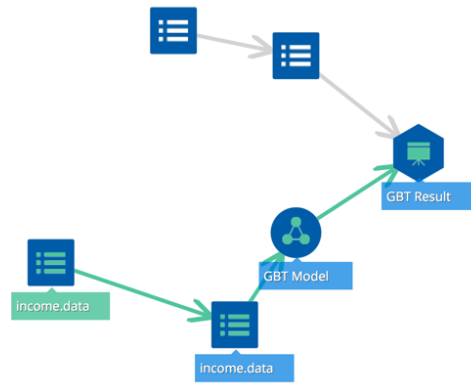
The following items will also be deleted:

income.data	DATASET
GBT Model	MODEL
GBT Result	TESTRESULT

View downstream objects in the **graph**

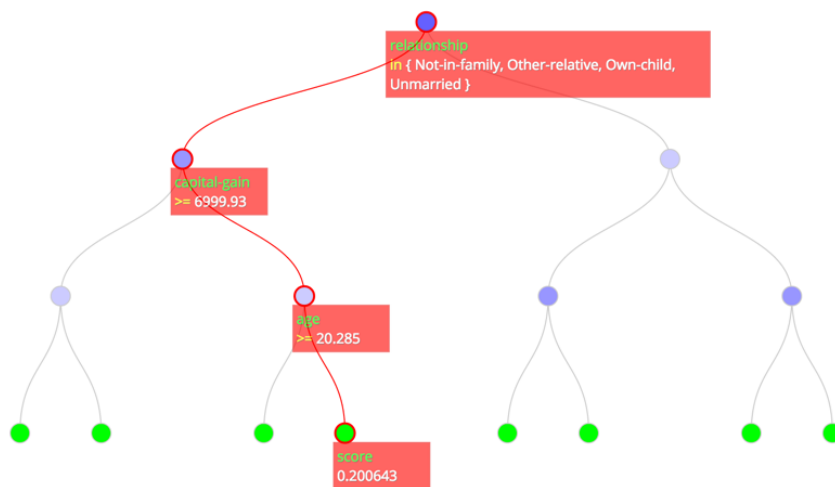
In this delete confirmation box, you can also click the **graph** link to view upstream and/or downstream files associated with this dataset.

☐ Display Upstream
☒ Display Downstream



Chapter 4 Models

A model is a tree-like representation of your dataset. In a model, each node corresponds to a feature field from a dataset. Except for the top node, each node has an incoming branch. Similarly, except for the bottom node (or leaf node), each node has a number of outgoing branches. A branch represents a possible value for the input field from the originating dataset. A leaf represents the value of the objective field, given all the values for each input field in the chain of branches that goes from the root to that leaf.



A model is trained on a dataset to predict a specific value. The Skytree GUI supports Random Decision Forest (RDF), Gradient Boosted Tree (GBT), Generalized Linear Models (GLM), and Support Vector Machine (SVM) models for classification. Random Decision Forest Regression, Gradient Boosted Tree Regression, and Generalized Linear Models (GLMR) are also supported for regression models. In addition, an AutoModel option is available for users who may be unsure of the parameters to include when tuning and training a model for predictions. AutoModel provides the convenience of performing an all-method smart search with minimal required parameter input. Users need only specify the training set and model validation options (tuning dataset, holdout ratio, number of folds, etc.), and AutoModel will automatically search for the best parameter setting without any additional required input from the user.

Adding Models

You create a model by selecting the Objective Column, which is the field you want to predict, as well as additional configuration options to use for tuning and training.

During the model creation process, the Skytree GUI automatically determines whether the problem you are trying to solve is classification or regression. This is based on whether the Objective Column is numeric or categorical.

Perform the following steps to add a model to a project:

1. From the Models page, click the **+ NEW MODEL** button.

Note: You can also create a new model when viewing a project's list of datasets. Select a dataset, then click **Create model** to start the model creation process.

2. Click the **Select** button to select the dataset that will be used to train the model. Note that the dataset list includes only those that have been added to the current project.
3. Specify a descriptive name for the model, such as Income Prediction.
4. Select the Objective Column. This represents the column to be predicted, such as, "yearly-income". Based on the selected Objective Column, the Skytree GUI automatically determines whether the problem to be solved is a classification or regression problem.
5. Select the machine learning method to be used to build the mode. In some cases, Skytree will suggest the correct method to use. For example, SVM and GLM/R might be more appropriate for models created from datasets that include non-categorical data. Available machine learning methods include:

Classification problems:

AutoModel, GBT, RDF, GLM, and SVM

Regression problems:

AutoModel, GBTR, RDRF, and GLMR

Note: Detailed information about the Skytree machine learning methods is available in the *Skytree Server User Guide*.

6. Select the parameters to use to build the model. The options vary based on the type of machine learning problem you are trying to solve and the machine learning method that you selected. In addition, the Skytree GUI will alert you if any dependencies exist when you specify an option. A message will appear beside the option, and an  icon on the related configuration sections.

Note: Refer to the [Command Reference](#) (page 47) section for information about the options available for each machine learning method.

7. Select the **Enable PMML** check box if you want to produce a PMML (Predictive Model Markup Language) output file.
8. As you select options for the model, Skytree will determine the resources required to build the model. The resources available in your YARN environment and the resources that will be used will display at the bottom of the screen. If not enough resources are available, Skytree will display the actual requirements, and the **Learn** button will be disabled. If resources are available, click **Learn** to build the model.

Dataset: Select

Model Name:

Objective Column: yearly-income ▾

ML Task: Classification

Method: GBT ▾

Smart search: ☐

Model validation type: None ▾

Method parameters: Show more ▾

Decision tree parameters: Show more ▾

Advanced method parameters: Show more ▾

Reproducibility seeds: Show more ▾

Enable PMML: ☐

Yarn Configuration

Number of nodes	3
Available memory for each node	50 GB
Available virtual cores for each node	16
Maximum memory for each container	50 GB
Maximum virtual cores for each container	32

Estimated Resource Requirements

Number of containers	1
Memory for each container	50 GB
Virtual cores for each container	16

Learn

The new model will appear immediately on the Models list. The Status column provides information about the status of the model. If the dataset used to create the model was not yet committed, the status will initially show ETL_INPROGRESS and indicate that the dataset is committing. When a dataset is committed, the status will show ML_INPROGRESS while the model is building. This column also provides details of the progress, such as the current number of trees built (for ensemble models), tuning parameter information (when tuning), and the percentage complete. The status will change to READY when the new model has been built.

LEAD SCORING - ADVANCED

Find me more new customers

DATASETS **MODELS** RESULTS PLOTS

+ New Model Search

LAST UPDATED	MODEL NAME	ID	TYPE	METHOD	STATUS
2016-02-15 10:44 AM	GBT Model	342038046773434755	Classification	GBT	ML_INPROGRESS - Overall Phases 0 of 2 (0%) - Parameter 0 of 3 (0%) - Tree 23 of 25 (92%) Started a few seconds ago
2016-02-15 10:30 AM	Income Prediction	7313134916943338074	Classification	GBT	READY

Models List

Models that are successfully added to a project will display in a model table. Click on a column heading to sort the data in that column. Mouse over an entry in the list to view the following additional options:

- **View tuning results:** Graphs the predicted value for the target field given a set of tuning parameters and a specified scoring method. Note that this option only displays for models that were built with tuning parameters.

- **View details:** Provides the configuration options used to create this model. If the model was tuned, then the tuning parameters will display as well.
- **View logs:** Review the log files that were created when the model was built. A log is available when the model is no longer INPROGRESS.
- **View graph:** View a graphical representation of this model beginning with the dataset that was used to create it. If this model was used to perform prediction and evaluation, then that will be represented as well.
- **Visualize:** View an interactive map showing the model tree structure and variable importances. Note that this option is only available for models that were built with GBT or GBTR.
- **Predict & Evaluate:** Using a model from this list, compare the model against an available dataset to predict and evaluate an ID column. Refer to [Adding Datasets](#) (page 9) for information on adding datasets.
- **Export PMML File:** Select this option to export PMML files to HDFS or downloaded to your local machine. This option is only available if the Enable PMML checkbox was selected when the model was built. In addition, this option is not available for distributed non-linear SVM models.
- **Delete:** Deletes the selected model.

LEAD SCORING - ADVANCED

Find me more new customers

DATASETS MODELS RESULTS PLOTS

+ New Model Search

LAST UPDATED	MODEL NAME	ID	TYPE	METHOD	STATUS
2015-12-04 10:28 AM	GBTR with Tuning	6510587813675571485	Regression	GBTR	READY
2015-12-01 11:48 AM	GBT with Tuning	7658272003427548845	Classification	GBT	READY
2015-12-01 11:45 AM	Automodel - Regre	5393390911128808004	Regression	Automodel (GBTR)	READY
2015-12-01 11:15 AM	Automodel - Classi	8270346448259536914	Classification	Automodel (GBT)	READY
2015-12-01 11:03 AM	GBTR Model	8931602808894388474	Regression	GBTR	READY
2015-12-01 11:02 AM	RDFR Model	6956086356737016276	Regression	RDFR	READY

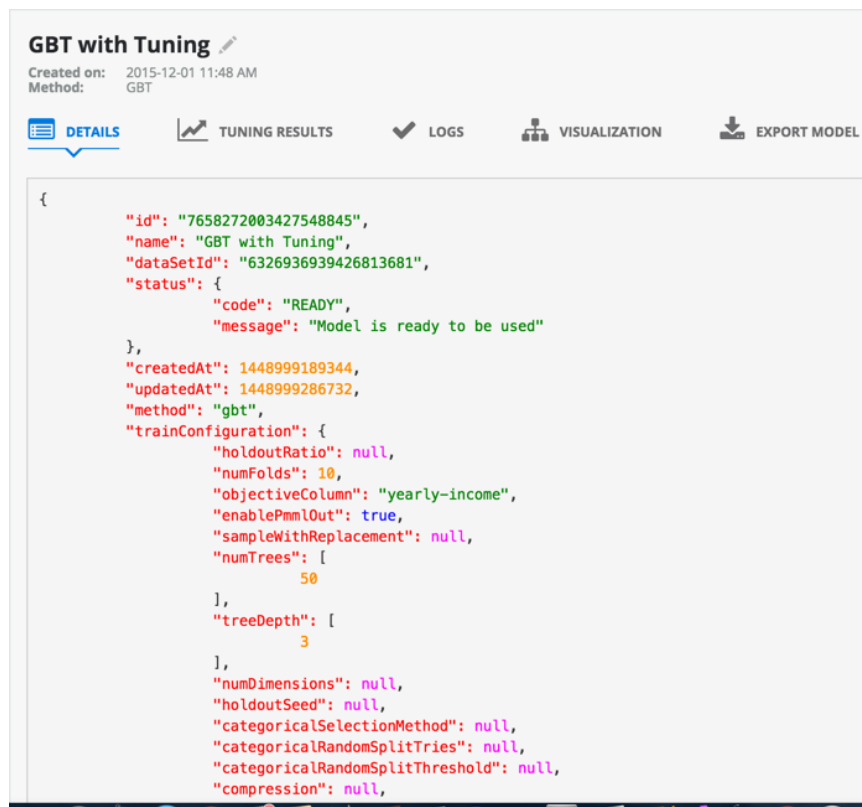
Most of these options are described in the sections that follow. Refer to [Predict & Evaluate](#) (page 36) in the Results chapter for information on how to perform a Predict and Evaluate operation.

View Model Details

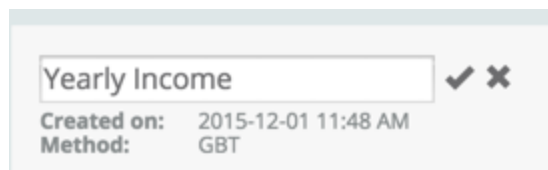
Select a model from the list, then click **View details** for more information about the selected model, including the ID, status, and configuration information. If tuning was performed, then this output also includes tuning results.

Click the links above the model details to navigate to the Tuning Results (if tuning was performed), Logs, Visualization (GBT and GBTR models only), and Export Model pages.

Note: The breadcrumbs on this page vary depending on the method used when building the model and whether tuning was performed. The Tuning Results option is only available for models that were built with tuning options. The Visualization option is only available for models built using GBT and GBTR. The Export Model option is only available if **Enable PMML** was selected when the model was built.



From this page, you can also edit the name of the model. Click the pencil icon (✎) beside the name, and then enter a new name for the model. Click the checkmark icon (✓) to complete the name change.

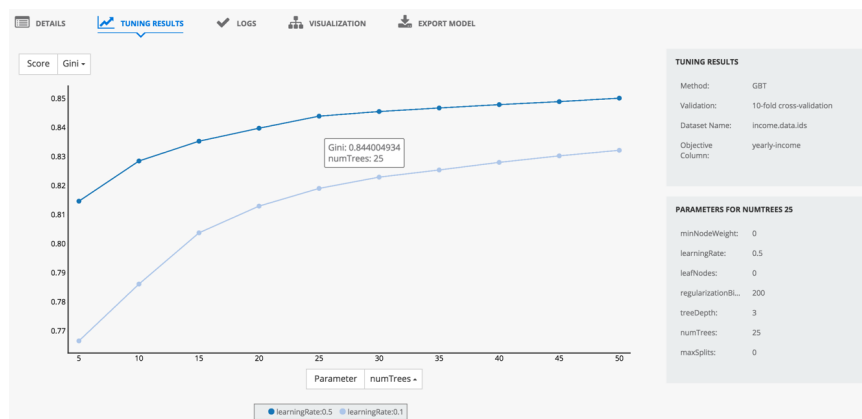


View Model Tuning Results

If a model has been tuned, then you can select that model from the list and click **View tuning results** to view a graphical display showing tuning results.

If the model was configured with a specific number of trees, then the output will show a graph with the number of trees (numTrees) on the X axis. Using dropdowns, the interactive display allows you to change the view of the graph by changing the Parameter variable on the X axis and the Score variable on the Y axis. Select a point on the graph to view more detailed information for the Score-Parameter combination.

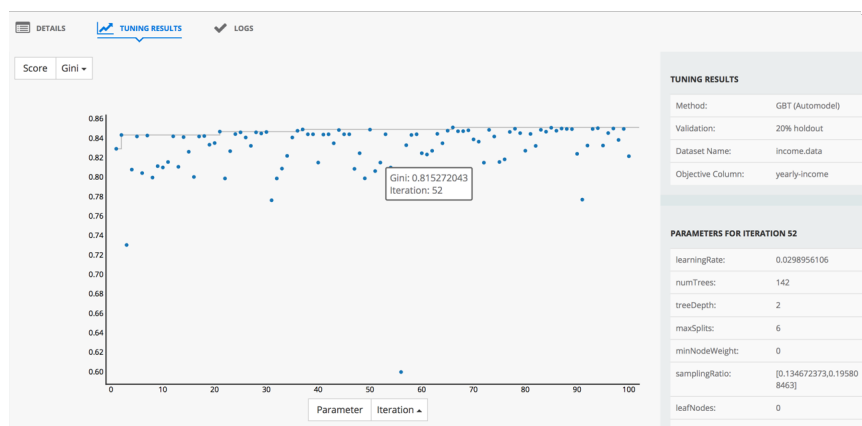
The following image shows a graph legend when a model is tuned using grid search. Items in the graph legend are sorted in descending order based on the best score (selected on the Y axis.) Click the links above the tuning results to navigate to the Model Details, Logs, and Visualization (GBT and GBTR models only) pages.



If tuning was performed using a specific number of iterations, as in the case of AutoModel, then the output by default shows a plot of the Gini scores for each iteration. You can mouse over points in the graph to show the Gini score for that point. In addition to mousing over, you can click on a point to view parameters that were used during tuning. Finally, you can change the scoring method using the Score drop down.

In the following image, AutoModel was run on the Income.data dataset using a 20% holdout set. When computing the results, AutoModel selected GBT as the method.

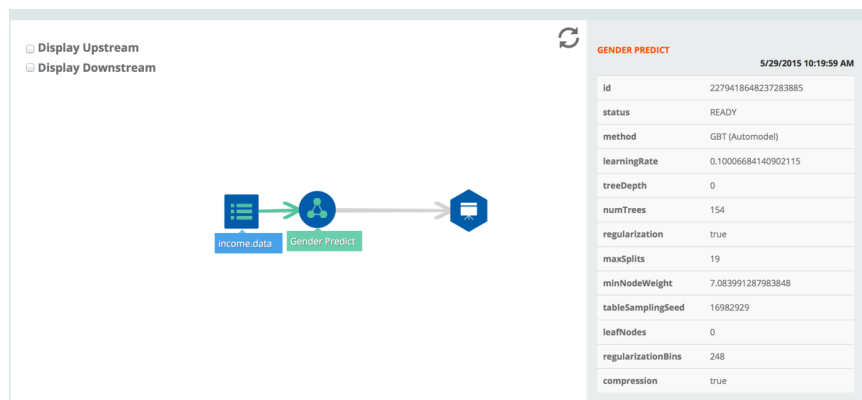
Note: The Visualization menu option is not available in the image below. This option is only available for models built using GBT and GBTR.



View Logs

Select a model from the list, then click the **View logs** button to view logging information that was generated when the model was trained. A log is available when the model is no longer INPROGRESS.

Click the links above the log to navigate to the Model Details, Tuning Results (if tuning was performed), Visualization (GBT and GBTR models only), and Export Model (if the Export PMML option was selected when building the model).



Visualize Models

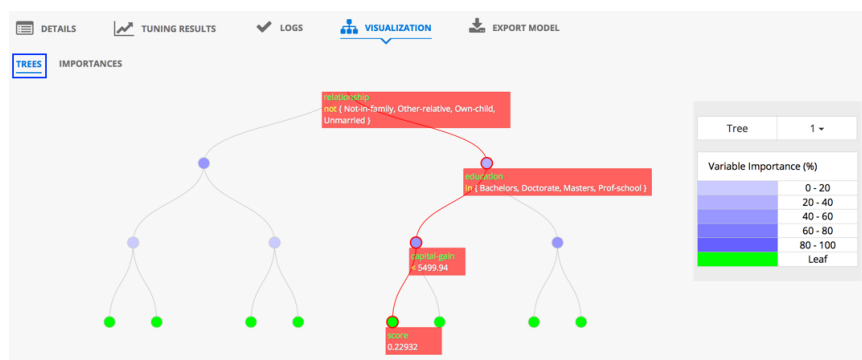
In the Skytree GUI, you can click **Visualize** for models that were built with GBT or GBTR. This opens an interactive map showing the model tree structure as well as a link to view the variable importances.

Click the links above the model visualization pages to navigate to the Model Details, Tuning Results (if tuning was performed), and Logs pages.

Model Visualization > Trees

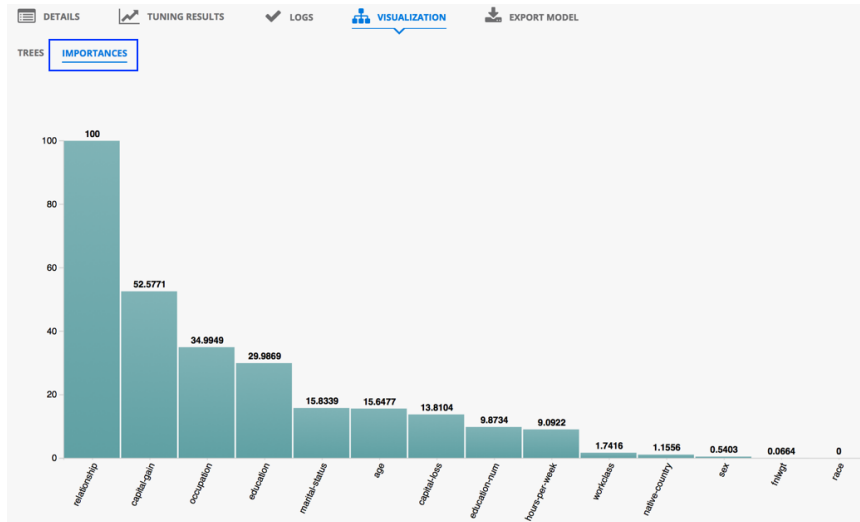
This page defaults to the **Trees** view, which provides a graphical representation of how different nodes are scored. Select different points on this graph to view paths and feature information for each leaf node and each score. If the model was built with multiple trees, then select the **Tree** dropdown on the right side to view graphical information for each tree that was built.

Skytree calculates the importance of each feature based on the whole model, and these importance scores are scaled so that the most important variable gets an importance of 100. The nodes of the tree are color coded based on the variable importance of the feature that was chosen to split the node. This variable importance percentage is computed once for the entire ensemble, and the coloring/shading is consistent across all trees.



Select the **Importances** link to view the relative importance of predictor variables in the model. This page provides a histogram of each feature in order of importance from most to least important.

Note: This page can display a maximum of 60 features.



Export PMML File

Trained models can be exported into PMML (Predictive Model Markup Language) format by selecting the **Enable PMML** checkbox when building the model. This allows the use of an external code for scoring. For more information on PMML format, please refer to the PMML 4.2 section on the [Data Mining Group Web site](http://www.dmg.org/v4-2/GeneralStructure.html) (<http://www.dmg.org/v4-2/GeneralStructure.html>).

Note: The **Enable PMML** check box is not available for distributed non-linear SVM models.

To access the PMML output:

1. Hover over the model that includes the PMML output, and select **Export PMML file**. This opens the Export Model tab.
2. On the Export Model tab, click the **Download Model** button.
3. Depending on your browser, you will be prompted to save the file locally or open it immediately using an XML viewer, or the file will download immediately to your local downloads folder.

Note: If the model export file is too large for your local machine, ask an Admin to download the file to HDFS.

Notes Regarding PMML Output

Please keep the following in mind when generating PMML output.

General

- For ensemble models, the size of the resulting PMML file is dependent on the number and depth of the trees. It may be very large.
- The PMML file that is produced requires that missing values are encoded as '?'. This is a reserved symbol for missing values, and as such, should not be used anywhere else in your data.

GBT/GBTR

In order to compute scores for the model, scores of all the trees in the model have to be summed. A function is then applied to the resulting sum. The function is provided under the node

```
<PostProcessor>
  <![CDATA[(1/(1 + e^(-2*s)))]>
</PostProcessor>
```

in the PMML file itself. The symbol *s* in the function refers to the sum of the scores of all the trees.

All options available for GBT are available for PMML output except for ranking functions (i.e., if the Loss Function is specified as other than `Logistic` (default)).

All options available for GBTR are available for PMML output.

Ensemble GBTs

Because Ensemble GBT applies the concept of “bagging”, the resulting models have sets of GBTs. Unfortunately, the standard PMML format does not currently support the concept of multiple models with a depth of hierarchy greater than one. In order to overcome this limitation we introduce custom PMML to correctly represent that model.

In the PMML export of Ensemble GBT, we embed a complete GBT within each `<Segment>` by introducing another level of `<Segmentation>`. Each GBT within a `<Segment>` block has its own `<PostProcessor>`, which is used to compute result of that ensemble. The results of all GBTs can then be averaged to get the final answer.

RDF/RDFR

The resulting model uses confidence rather than probability to express each leaf’s `ScoreDistribution` elements. Confidence is different from probability in that the confidences of all classes do not necessarily sum to one, and in fact nodes can have confidence values that are greater than one.

Given a query point, prediction is performed by, for each class, summing the confidences at the leaf nodes encountered by the query point in all the trees. Then, the posterior probability of a given class is found by dividing that class’s total confidence by the sum of all classes’ confidences. Alternately, the class with the largest overall confidence may be taken as the query point’s predicted class.

All options available for RDF and RDFR are available for PMML output.

Delete Models

Select an entry from the list of models, then click **Delete** to delete the model along with all downstream files that were generated from this model (for example, test results and plots).

When you click **Delete**, Skytree will prompt you with to confirm. This confirmation box will include list of all files that the model was referenced in. Click **Delete** again to delete the model and all associated files.

Delete Model?



Do you want to delete **GBT Model**?

The following items will also be deleted:

GBT Result

TESTRESULT

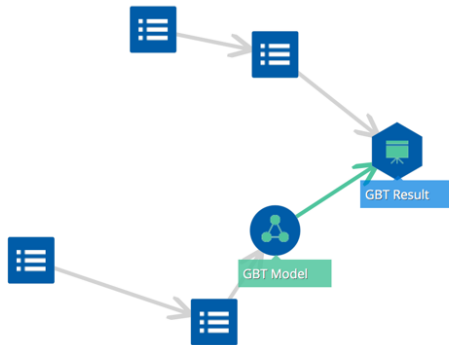
View downstream objects in the **graph**

Delete

Cancel

In this delete confirmation box, you can also click the **graph** link to view upstream and/or downstream files associated with this model.

- ☐ Display Upstream
☒ Display Downstream



Chapter 5 Results

The Results page provides a list of computed results in a project.

Classification Scoring

For classification methods, predicted probabilities and labels can be compared against known results. The Skytree classification methods can be used to compute the confusion matrix and derived metrics, Gini, and capture deviation.

- **Confusion Matrix**

In classification scoring, the confusion matrix is a table to visualize the performance of certain supervised machine learning methods, such as classifiers. Each column of the matrix represents the instances in a predicted class, while each row represents the instances in an actual class. For binary classifiers, the confusion matrix summarizes the number of correct and false classifications in both directions (i.e., both true and false positives and negatives).

The confusion matrix is also available with a precision recall (PR) curve. The output includes the precision, recall, probability threshold, true positives (TP), false positives (FP), true negatives (TN), false negatives (FN), and the percentile (number of points vs. total number of points).

- **Gini Index**

The Gini index is a well-established method to quantify the inequality among values of a frequency distribution, and can be used to measure the quality of a binary classifier. A Gini index of zero expresses perfect equality (or a totally useless classifier), while a Gini index of one expresses maximal inequality (or a perfect classifier).

The Gini index is defined based on either the receiver operating characteristic (ROC) curve or the Lorenz curve. The ROC curve plots the true positive rate (y-axis) as a function of the false positive rate (x-axis), while the Lorenz curve plots the true positive rate (y-axis) as a function of percentiles of the population (x-axis).

- **Capture Deviation**

Capture deviation is a normalized sum of piece-wise absolute deviations between the predicted probabilities and the actual probabilities. The best possible capture deviation value is 0. The actual and predicted probabilities are obtained by counting the actual fraction of points in class 1, and by taking the average over the predicted probabilities to be in class 1 respectively.

Regression Scoring

For regression methods, predicted regression targets can be compared against testing targets, and typical error metrics such as mean absolute error, mean squared error, root mean squared error, and L1/L2 error are meaningful

quantities to consider. Skytree calculates all of the above, in addition to the Coefficient of Determination and Normalized Gini.

Note: Normalized Gini is ill-defined for datasets with any negative response values. If this scenario is detected, the normalized Gini is calculated with all response values shifted by $-t_{min}$, where t_{min} is the minimum response value. This creates a non-negative set of response values.

Predict & Evaluate

Perform the following steps to generate a new prediction file:

1. From the Results page, click the **+ New Predict & Evaluate** button.
2. Select the Model to be used as the basis for the prediction.
3. Enter a name for the generated results.
4. Select the Dataset that will be compared against the model.

Note: Only datasets that were imported with an ID column can be specified.

5. Click **Predict & Evaluate** when you are ready.

Model: GBT with Tuning [Select]

Method: GBT

Objective column: yearly-income

Result name: Yearly Income Prediction

Dataset: income.test.ids [Select]

ID column: id

[Predict & Evaluate]

The new result will appear immediately on the Results list. The Status column show In Progress while the result is building. This will change to READY when the new result is available, and the Test Scores column will populate with the results.

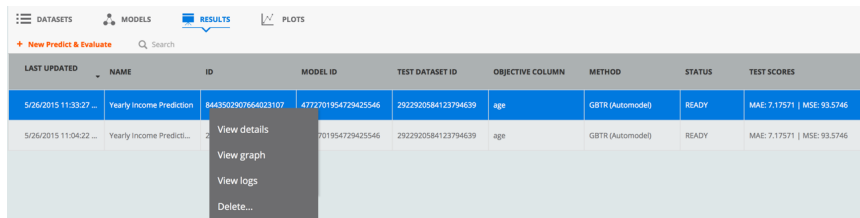
LAST UPDATED	NAME	ID	MODEL ID	TEST DATASET ID	OBJECTIVE COLUMN	METHOD	STATUS	TEST SCORES
2015-12-02 10:45 AM	Income Prediction	1843092245051501927	7658272003427548845	2126943090296487771	yearly-income	GBT	READY	Gini: 0.8737 Accuracy: 0.8704

Results List

This summary page shows the Model and Test Dataset IDs that were used during the test, the target column, the machine learning method, and the test scores. Click on a column heading to sort the data in that column. Mouse over

an entry in the list to view the following additional options:

- **View details:** View additional information about the test, including a results summary, ROC curve, confusion matrix, and scores. Note that the information that displays can vary depending on the machine learning method used to produce the result. Similarly, if you performed a Predict only, then scoring information (including confusion matrix and ROC curve) will not be available.
- **View graph:** View a graphical representation of this test result's history, from importing the original dataset through testing.
- **View logs:** Review the log files that were created when the result was generated. A log is available when the result is no longer INPROGRESS.
- **Delete:** Deletes the selected result.



The screenshot shows the 'RESULTS' tab in the Skytree GUI. A table lists test results with columns: LAST UPDATED, NAME, ID, MODEL ID, TEST DATASET ID, OBJECTIVE COLUMN, METHOD, STATUS, and TEST SCORES. Two rows are visible, both for 'Yearly Income Prediction' using 'GBTR (Automodel)'. A context menu is open over the first row, showing options: View details, View graph, View logs, and Delete...

LAST UPDATED	NAME	ID	MODEL ID	TEST DATASET ID	OBJECTIVE COLUMN	METHOD	STATUS	TEST SCORES
5/26/2015 11:33:27 ...	Yearly Income Prediction	8443502907664023107	4772701954729425546	2922920584123794639	age	GBTR (Automodel)	READY	MAE: 7.17571 MSE: 93.5746
5/26/2015 11:04:22 ...	Yearly Income Prediction...	...	701954729425546	2922920584123794639	age	GBTR (Automodel)	READY	MAE: 7.17571 MSE: 93.5746

View Result Details

Select a test result from the Results page, then select **View details** for information about the test results. The information that displays on this page can vary depending on the machine learning method used when the model was built.

The Results page is separated into the following sections:

- **Result Information**

This includes a summary of the data used when producing the result. This sections shows the model name and ID, the machine learning method used to generate the result, the test dataset name and ID used to compare against the model, and the status of the test result.

- **Confusion Matrix**

For classification scoring results, the confusion matrix section displays the outcome of certain supervised machine learning methods, such as classifiers. Each column of the matrix represents the instances in a predicted class (Predicted Positives and Predicted Negatives). Each row represents the instances in an actual class (Actual Positives and Actual Negatives). For binary classifiers, the confusion matrix summarizes the number of correct and false classifications in both directions (i.e., both true and false positives and negatives).

Note: This information is not available in Regression scoring.

- **Cost/Benefit Weighted Evaluation**

For classification scoring results, values for true positives/negatives default to 1, and values for false positives/negatives default to -1. Enable this checkbox to open the Cost/Benefit Weighted Evaluation form and view results based on different values. As you enter new values, the Confusion Matrix updates showing the updated predicted and actual values. The weighted values are displayed in parentheses. Below the form, the Cost/Benefit value recalculates based on the updated values.

CONFUSION MATRIX

n = 9778

	Predicted Positive	Predicted Negative
Actual Positive	1833 (2200)	543 (652)
Actual Negative	809 (971)	6593 (7912)

COST/BENEFIT WEIGHTED EVALUATION ☒ Enable

True Positive Benefit

1.2

True Negative Benefit

1.2

False Positive Cost

-1.2

False Negative Cost

-1.2

Cost/Benefit Value

8489

- Scores**

For classification methods, predicted probabilities and labels are compared against known results. Skytree computes the Accuracy, Recall, False Positive Rate, Precision, F-Score, Gini, and Capture Deviation.

For regression methods, predicted regression targets are compared against testing targets and typical error metrics such as Mean Absolute Error, Mean Squared Error, Root Mean Squared Error, and L1/L2 error. Skytree calculates each of these in addition to the Coefficient of Determination and Normalized Gini.

- Test Set Scores**

For classification results, this section provides options for downloading the result Probabilities and Labels files to your local machine.

TEST SET SCORES

Labels:

Download Labels

Probabilities:

Download Probabilities

For regression results, this section provides options for downloading the result Target file to your local machine.

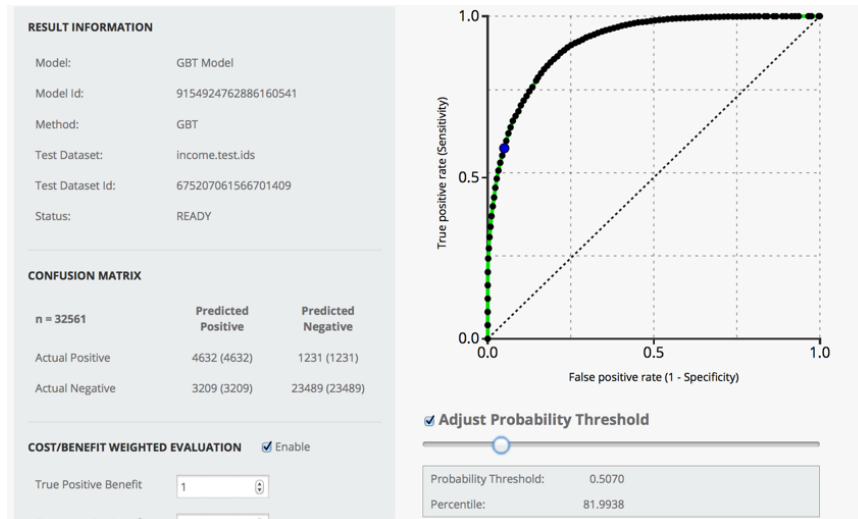
TEST SET SCORES

Targets:

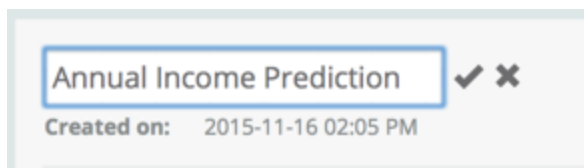
Download Targets

- **ROC Curve**

For classification results, a graph displays showing the ROC curve. By default, this graph shows a probability threshold of 1.0000. Users can enable the Adjust Probability Threshold option to select different points on the curve and view the actual and predicted true and false positives/negatives for those points.



From this page, you can edit the name of the result. Click the pencil icon (✎) beside the name, and then enter a new name for the result. Click the checkmark icon (✓) to complete the name change.



View Graph

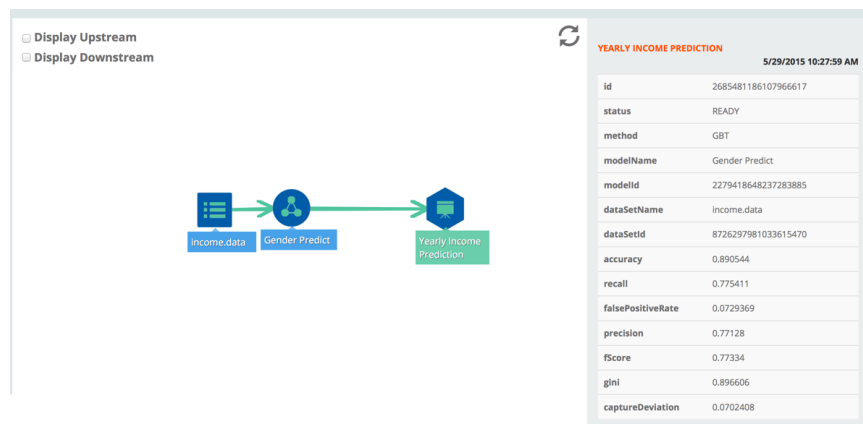
Select a test result from the list, then click **View graph** to see graphical view of the files in this project. This provides a complete lifecycle of the result beginning with the original dataset.

A test result lifecycle can include the following information:

	This represents a dataset.
	This represents a model built from a dataset.
	This represents a Predict & Evaluate result.

The graphical information is interactive. When you select a result in the graph, additional information about the result displays in the right column, including the model and dataset that were used to generate the result and the scoring information. In the graph below, you can see that the `income.data` file was used to build the Gender Predict model.

These were both used to generate a new test result called Yearly Income Prediction. The right side of the graph includes the scoring information. You can click on the title in the right section to view the result details. Use the **Display Upstream** and **Display Downstream** options to view the specific files that are related to this result.

[View Logs](#)

Select a result from the list, then click the **View logs** button to view logging information that was generated when the result was created. A log is available when the result is no longer INPROGRESS.

```
skytree (X) - THE MACHINE LEARNING COMPANY (TM) - http://www.skytree.net/
(C) 2018-2019 Skytree Inc. All rights reserved.
=====
Release: develop
Source ID: libanlr4
Source Date: Fri May 22 17:36:24 2015 -0700
Compiler: 6.5.0
Compile Date: May 22 2015 18:05:00
=====
Local User: 2015-May-20 14:33:18
Hostname: c8330b04.skytree.io
System: Red Hat Enterprise Linux Server release 6.5 (Santiago)
Processor: Intel(R) Xeon(R) CPU E5-2670 v2 @ 2.90GHz
# CPU Cores: 16 (Hyperthreading)
Total Memory: 128.31 GB
Free Memory: 131.36 GB
System Load: 1.12 %
=====
Working directory: /home
Command Line arguments:
skytree-server thrift --port 9000,9001,9004,9005,9006,9007,9008,9009,9010 --master host=c8330b04.skytree.io --master-port=9009 --session_id=3862255373884929c1e75f7_64135d207604d2107

14:31:11 [INFO] Module specified: thrift
14:31:11 [INFO] Running in non-distributed mode.
14:31:11 [INFO] System status:
14:31:11 [INFO] |-----|
14:31:11 [INFO] |      | Avail. | Avail. | System | Total | Free |
14:31:11 [INFO] | # | Hostname | Cores | Threads | Load (%) | Mem. (GB) | Mem. (GB) |
14:31:11 [INFO] |-----|-----|
14:31:11 [INFO] | 0 | c8330b04.skytree.io | 8 | 16 | 1.12 | 128.31 | 131.37 |
14:31:11 [INFO] |-----|-----|
14:31:11 [INFO] Full hostnames:
14:31:11 [INFO] -----
14:31:11 [INFO] 0 | c8330b04.skytree.io |
14:31:11 [INFO] -----
14:31:11 [INFO] Starting master server at port 9001
14:31:14 [INFO] Server started
14:31:14 [INFO] Saving master server data from /user/skytree/skytree/data_logs/3862255373884929c1e75f764135d207604d2107/logs/3862255373884929c1e75f764135d207604d2107 - testing in /user/skytree/skytree/data_logs/3862255373884929c1e75f764135d207604d2107/data/data_logs/3862255373884929c1e75f764135d207604d2107
14:31:17 [INFO] Loading testing data from /user/skytree/skytree/data_logs/3862255373884929c1e75f764135d207604d2107/data/data_logs/3862255373884929c1e75f764135d207604d2107
```

Delete Results

Select an entry from the list of results, then click **Delete** to delete the result along with all downstream files that were generated from this results (for example, plots and datasets).

When you click **Delete**, Skytree will prompt you with to confirm. This confirmation box will include list of all files that the result was referenced in. Click **Delete** again to delete the result and all associated files.

Delete Result?



Do you want to delete **Yearly Income Prediction**?

The following items will also be deleted:

test_result_source_1133719699457840027	SOURCE
test_result_dataset_1133719699457840027	DATASET
test_result_dataset_1133719699457840027 ...	DATASET
test_probabilities_source_113371969945784...	SOURCE
test_result_dataset_1133719699457840027	DATASET
GBT Result plot	PLOT

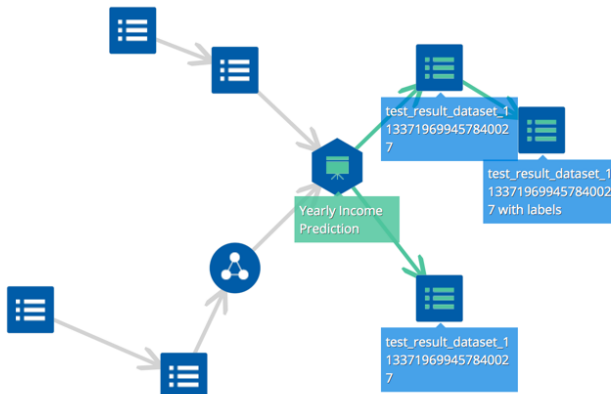
View downstream objects in the [graph](#)

Delete

Cancel

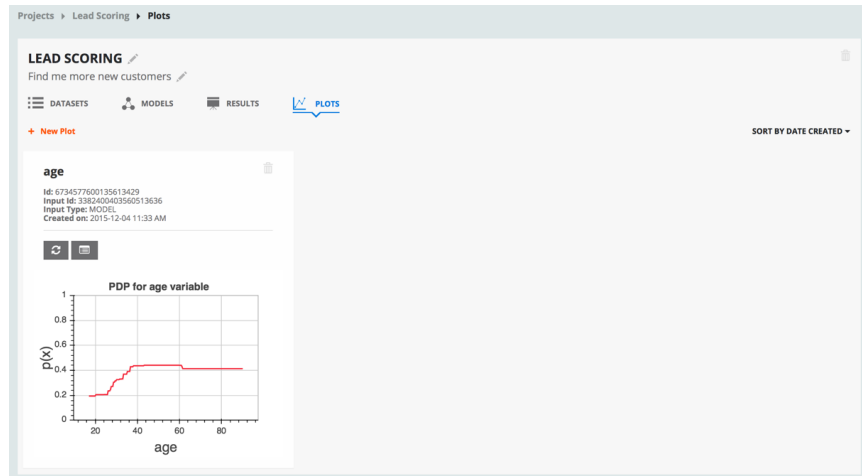
In this delete confirmation box, you can also click the **graph** link to view upstream and/or downstream files associated with this result.

- ☐ Display Upstream
☒ Display Downstream



Chapter 6 Plots

The Skytree GUI allows you to create plots to see a graphical view of your data. A plot is based off of either a model or a test result, depending on the data that you want to view. When used with models, these plots show the dependence between a feature or set of features marginalized over all other features. When used with results, the plots show either the relation between predictors and actual outcomes (regression) or the capture deviation information (classification).



These plots are interactive. Using your mouse, you can increase the view, decrease the view, or adjust the value display within a graph. Use the reset () button to return the plot data to the default view.

You can view a larger display of a single plot by selecting the details () button. Use the breadcrumbs at the top of the Details page to return to the Plots page.

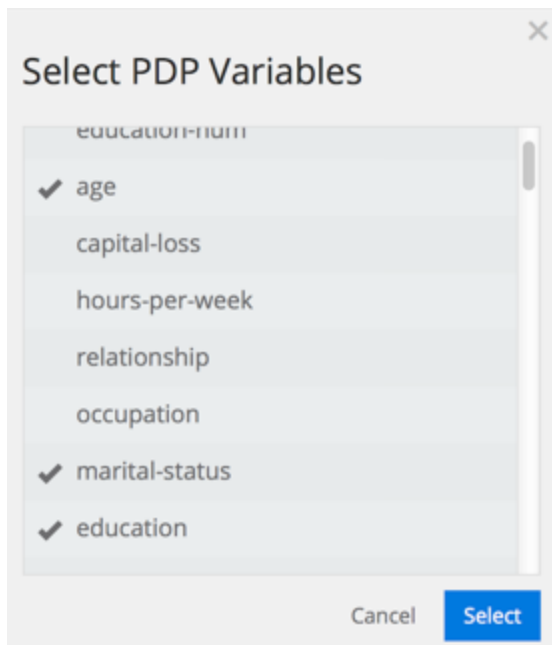
The Plots page also includes a sort drop-down menu. When multiple plots appear on a page, use this menu to arrange the plots based on the plot name, the date created, the plot type (model or result), or the plot input id (the model ID or result ID from which a plot is based).



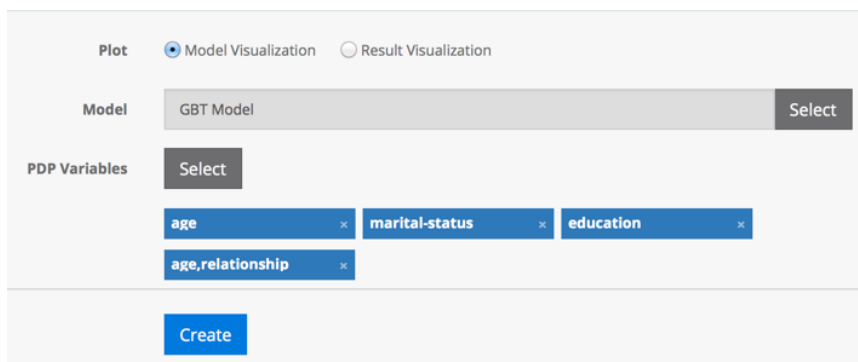
Adding a Model Visualization Plot

Perform the following steps to add a model visualization plot to a project:

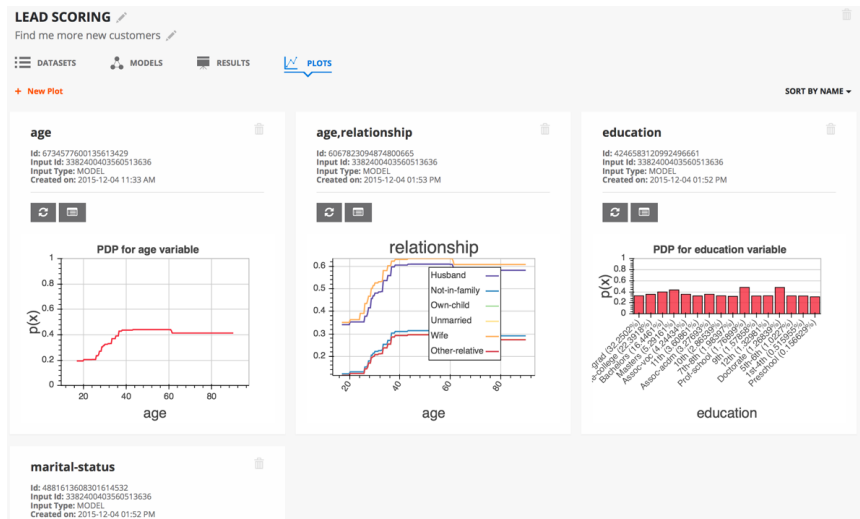
1. From the Plots page, click the **+ New Plot** button.
2. The **Model Visualization** option is selected by default. Specify the model that you want to use for the plot.
3. Select one or more PDP variables or variable pairs to include in the plot, then click **Select**. A plot will be created for each selected variable.



4. Click **Create** to build the plot.



The newly created plots will display on the Plots page.



Adding a Result Visualization Plot

Perform the following steps to add a model visualization plot to a project:

1. From the Plots page, click the **+ New Plot** button.
2. Select the **Result Visualization** option.
3. Model visualization plots are named based on the selected PDP variable. For Result visualization plots, you must enter a name for the plot.
4. Specify the result that you want to use for the plot.
5. For Classification result plots, you can specify of the number of buckets to include in the capture deviation plot. Capture deviation is a normalized sum of piece-wise absolute deviations between the predicted probabilities and the actual probabilities. The best possible capture deviation value is 0. The actual and predicted probabilities are obtained by counting the actual fraction of points in class 1, and by taking the average over the predicted probabilities to be in class 1 respectively. The numBuckets value defaults to 10, leading to the averages being computed within the deciles of the predicted probability.
6. Click **Create** to build the plot.

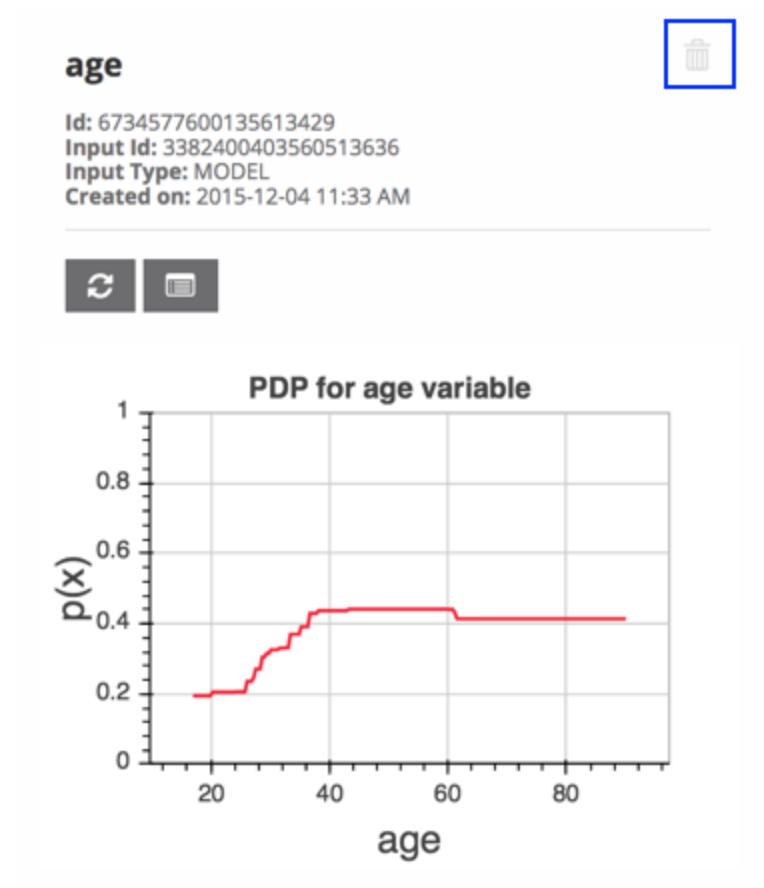
The form for creating a Result Visualization plot includes the following fields and options:

- Plot**: Radio buttons for **Model Visualization** and **Result Visualization** (selected).
- Plot Name**: Text input field containing 'Result Visualization - Classification'.
- Result**: Dropdown menu showing 'Yearly Income Prediction' with a **Select** button.
- Summarizer Properties**: Section containing a **numBuckets** input field set to '10'.
- Create**: A blue button at the bottom to build the plot.

The newly created plot will display on the Plots page.

Deleting Plots

You can delete a plot by clicking the delete icon (🗑️) in the upper-right corner of the plot.



After clicking the delete icon, you will be prompted to continue with the deletion. Click **Delete** to continue or **Cancel** to return to the Plots page without deleting the plot.

Delete? ×

Do you want to delete **age**?

Delete Cancel

Appendix A Command Reference

This appendix describes the commands that are available for use with the Skytree GUI based on the selected method. Refer to the following sections for additional information:

- [AutoModel Options](#) (page 47)
- [GBT Options](#) (page 49)
- [GBTR Options](#) (page 51)
- [GLM Options](#) (page 54)
- [GLMR Options](#) (page 56)
- [RDF Options](#) (page 58)
- [RDFR Options](#) (page 61)
- [SVM Options](#) (page 62)

AutoModel Options

The following options are available when building AutoModel models.

Table 1: AutoModel Parameters

Parameter	Description
Search iterations	Specify the number of search rounds to try for tuning. This value must be greater than 0 and defaults to 100.
Restrict tuning parameter space	This option restricts the parameter space (for example, by limiting the number of trees, tree depth, etc.) in order to reduce the training time. This option is enabled by default. Note that when GBT or GBTR is the method used to generate the model, then the following upper bound values will be set: • Number of trees ≤ 3000 • Tree depth ≤ 9 • Maximum splits ≤ 256

Table 2: Model Validation - AutoModel

Parameter	Description
Number of folds (k)	Optionally specify the number of folds for k-fold cross-validation. If a Holdout ratio is also specified, then Monte Carlo cross-validation (with randomly drawn holdout sets) is performed with the specified number of restarts.
Holdout ratio	Specify the fraction of the table to be held out for tuning. This value defaults to 0.2.
Classifier testing objective	When tuning a classification problem, specify the objective for selecting the test model from a set of tuned models. Options include: • Gini (default) • F-Score • Accuracy • Capture deviation • Precision@k • Yield
Classification objective	When tuning a classification problem, specify the objective for the classification threshold. Options are F-Score (default) and Accuracy.
Regressor testing objective	When tuning a regression problem, specify the objective for selecting the test model from a set of tuned models. Options include: • Mean absolute error (default) • Mean squared error • Coefficient of determination • Normalized Gini
k for Precision@k	Optionally specify <i>k</i> as a value between 0 and 1 for precision at the top 100k-th percentile. <i>k</i> should be greater than 0 and less than 1. This value defaults to 0.1.

Table 3: Reproducibility seeds - AutoModel

Parameter	Description
Model validation sampling seed	Optionally specify the model validation sampling seed. This serves as the holdout sampling random number seed and/or the seed for the number of folds. If omitted, a time-based seed will be used
Smart search sampling seed	Optionally specified a seed value for smart search. If omitted, a time-based seed will be used by default.

Table 4: PMML output - AutoModel

Parameter	Description
Enable PMML	Optionally select this checkbox if you want to produce PMML (Predictive Model Markup Language) output.

GBT Options

The following options are available when building GBT models.

Table 5: Smart search - GBT

Parameter	Description
Smart search	Specify to use an intelligent tuning technique (instead of naive grid search). If specified, then only model validation options must be specified (Number of folds and/or Holdout ratio). Users have the option to specify additional tuning parameters, but this is not required. Skytree will determine the best parameter settings to use during tuning.
Search iterations	If Smart search is enabled, then optionally specify the number of search rounds to try for tuning. This value must be greater than 0 and defaults to 100.

Table 6: Model validation type options - GBT

Parameter	Description
Number of folds (k)	Optionally specify the number of folds for k-fold cross-validation.
Holdout ratio	Optionally specify the fraction of the table to be held out for tuning.
Monte Carlo cross-validation	Monte Carlo cross-validation can be used specifying a Holdout ratio in conjunction with the number of folds. With Monte Carlo cross validation, for example, if the number of folds is 10 and the holdout ratio is 0.2, then a random 20% of the training data would be heldout in each fold as the validation set.

Table 7: Method parameters - GBT. Tunable options can be specified as a single value, a comma separated list, or as an array.

Parameter	Description
Loss function	Specify the method to use when performing ranking for non-ensemble GBT. Options include the following: Logistic (default)
Number of trees	Specify a list of numbers to indicate the number of trees in the ensemble. This is a tunable option and can be specified a single value or as an array. This value must be > 0.

Table 7: Method parameters - GBT. Tunable options can be specified as a single value, a comma separated list, or as an array. (continued)

Parameter	Description
Learning rate	Specify a list of decimal values value for the learning rate for each iteration. This is a tunable option and can be specified a single value or as an array. This value defaults to 0.1 and must be >0 and <=1.
Sampling ratio	Specify a value to indicate the per-class sampling ratio. You can use a single Sampling Ratio that will be used for each class, or specify Sampling Ratio values for each class. If a value is not specified, then full bootstrapping will be used.
Sample with replacement	Specifies that for each tree, sample the training points using replacement (bootstrap). This option is on by default. If turned off, then a Sampling ratio value must be provided.
Imbalance	If this check box is enabled, then Skytree will attempt to improve classification for imbalanced classes.
Imbalance scale	If Imbalance is enabled, then specify a list of decimal values for the relative size of all classes with respect to the smallest class. This is a tunable option and can be specified a single value or as an array.

Table 8: Decision tree parameters - GBT. Tunable options can be specified as a single value, a comma separated list, or as an array.

Parameter	Description
Maximum number of splits	Specify the number of splits in the tree. This value must be >=0. A value of 0 disables this option. Either this value or Minimum node weight must be >0 if Tree depth is explicitly set to 0. If both Tree depth and Maximum number of splits are >0, then Skytree finds the best splits that exist within the depth level, up to Maximum number of splits.
Tree depth	Specify a list of integers for the depth to which each ensemble is built. If set to 0, trees will be built to the fullest extent. In this case, then you must also specify a positive value for either Maximum number of splits or Minimum node weight. If this value is not set, and values for Maximum number of splits and Minimum node weight are likewise not set, then this value defaults to 3. If this value and Maximum number of splits are both >0, then Skytree finds the best splits that exist within the depth level, up to Maximum number of splits.
Minimum node weight	Specify the minimum bound on the total weight in each leaf during tree building. This value defaults to 0. Either this value or Maximum number of splits must be >0 if Tree depth is explicitly set to 0.
Regularization	This option tells Skytree to use the fast decision tree construction heuristic. Multiple nodes can be used when this option is enabled without the need for additional tuning parameters.

Table 8: Decision tree parameters - GBT. Tunable options can be specified as a single value, a comma separated list, or as an array. (continued)

Parameter	Description
Regularization bins	Specify a list of integers for the number of bins to use per dimension/attribute of the data. This option is only used with Regularization. If Regularization is enabled, then this value defaults to 200.

Table 9: Advanced method parameters - GBT

Parameter	Description
Ensemble size	Specify a list of integers. The ensemble GBT algorithm applies the concept of bootstrap aggregation, or “bagging,” to GBTs. The training table is sampled a number of times, and a new GBT is built for each sample. This option specifies the number of GBTs to build.
Trim	If enabled, observations with low importance in the model will be pruned in successive iterations. This can lead to significant speedup, but accuracy can suffer. This option defaults to off.
Trim alpha	If Trim is enabled, then specify a decimal value > 0 and <1 as the alpha value. Higher values lead to more aggressive pruning of observations per iterations.

Table 10: Reproducibility seeds - GBT

Parameter	Description
Table sampling seed	Specify a long value for the data sampling random number seed. If omitted, a time-based seed will be used.
Dimension sampling Seed	Specify a long value for the dimension sampling random number seed. If omitted, a time-based seed will be used.
Model validation sampling seed	Optionally specify the model validation sampling seed. This serves as the holdout sampling random number seed and/or the seed for the number of folds. If omitted, a time-based seed will be used
Smart search sampling seed	Optionally specified a seed value for smart search. If omitted, a time-based seed will be used by default.

Table 11: PMML output - GBT

Parameter	Description
Enable PMML	Optionally select this checkbox if you want to produce PMML (Predictive Model Markup Language) output.

GBTR Options

The following options are available when building GBTR models.

Table 12: Smart search - GBTR

Parameter	Description
Smart search	Specify to use an intelligent tuning technique (instead of naive grid search). If specified, then only model validation options must be specified (Number of folds and/or Holdout ratio). Users have the option to specify additional tuning parameters, but this is not required. Skytree will determine the best parameter settings to use during tuning.
Search iterations	If Smart search is enabled, then optionally specify the number of search rounds to try for tuning. This value must be greater than 0 and defaults to 100.

Table 13: Model validation type options - GBTR

Parameter	Description
Number of folds (k)	Optionally specify the number of folds for k-fold cross-validation.
Holdout ratio	Optionally specify the fraction of the table to be held out for tuning.
Monte Carlo cross-validation	Monte Carlo cross-validation can be used specifying a Holdout ratio in conjunction with the number of folds. With Monte Carlo cross validation, for example, if the number of folds is 10 and the holdout ratio is 0.2, then a random 20% of the training data would be heldout in each fold as the validation set.

Table 14: Method parameters - GBTR. Tunable options can be specified as a single value, a comma separated list, or as an array.

Parameter	Description
Loss Function	Optionally specify a loss function method to use during regression. Available values include the following: ▪ Least absolute deviance- Perform least absolute deviation (LAD) regression ▪ Least Squares - Perform least-squares regression ▪ Huber - Perform Huber loss regression ▪ Poisson (log) - Perform Poisson-log regression with positive integral targets (such as counts) ▪ Gamma (log) - Perform Gamma-log regression with positive continuous targets ▪ Gamma (inverse)- Perform Gamma-inverse regression with positive continuous targets ▪ Tweedie (log) - Perform Tweedie-log regression If Huber is specified, then you must also define a value for Huber Loss Quantile. If Poisson (log) is specified, then a Log clamp value is required. Finally, if Tweedie (log) is specified, then you must also define a value for the Tweedie exponent.
Number of trees	Specify a list of numbers to indicate the number of trees in the ensemble. This is a tunable option and can be specified a single value or as an array. This value must be > 0.

Table 14: Method parameters - GBTR. Tunable options can be specified as a single value, a comma separated list, or as an array. (continued)

Parameter	Description
Learning rate	Specify a list of decimal values value for the learning rate for each iteration. This is a tunable option and can be specified a single value or as an array. This value defaults to 0.1 and must be > 0 and <=1.
Sampling ratio	Specify a list of doubles to indicate the per-class sampling ratio. You can use a single Sampling Ratio that will be used for each class, or specify Sampling Ratio values for each class. If a value is not specified, then full bootstrapping will be used.
Sample with replacement	Specifies that for each tree, sample the training points using replacement (bootstrap). This option is on by default. If turned off, then a Sampling ratio value must be provided.
Huber loss quantile	When Loss function is Huber, specify the top percentile of error that should be considered as outliers. This value must be between 0 and 1.
Log clamp	When the Loss function is configured as is Poisson (log), Gamma (log), or Tweedie (log), specify the maximum absolute log value for any terminal node. This value must be > 0.
Tweedie exponent	When a Tweedie loss function is specified, then you must also specify a value for the exponent. This value must be > 1.0 and < 2.0. If users desire a value of 1, they should use Poisson distribution. Similarly, if users desire an exponent equal to 2.0, they should specify Gamma distribution.

Table 15: Decision tree parameters - GBTR. Tunable options can be specified as a single value, a comma separated list, or as an array.

Maximum number of splits	Specify the number of splits in the tree. This value must be >=0. A value of 0 disables this option. Either this value or Minimum node weight must be > 0 if Tree depth is explicitly set to 0. If both Tree depth and Maximum number of splits are > 0, then Skytree finds the best splits that exist within the depth level, up to Maximum number of splits.
Tree depth	Specify a list of integers for the depth to which each ensemble is built. If set to 0, trees will be built to the fullest extent. In this case, then you must also specify a positive value for either Maximum number of splits or Minimum node weight. If this value is not set, and values for Maximum number of splits and Minimum node weight are likewise not set, then this value defaults to 3. If this value and Maximum number of splits are both > 0, then Skytree finds the best splits that exist within the depth level, up to Maximum number of splits.
Minimum node weight	Specify the minimum bound on the total weight in each leaf during tree building. This value defaults to 0. Either this value or Maximum number of splits must be > 0 if Tree depth is explicitly set to 0.

Table 15: Decision tree parameters - GBTR. Tunable options can be specified as a single value, a comma separated list, or as an array. (continued)

Regularization	This option tells Skytree to use the fast decision tree construction heuristic. Multiple nodes can be used when this option is enabled without the need for additional tuning parameters.
Regularization bins	Specify a list of integers for the number of bins to use per dimension/attribute of the data. This option is only used with <code>Regularization</code> . If <code>Regularization</code> is enabled, then this value defaults to 200.

Table 16: Advanced method parameters - GBTR

Parameter	Description
Ensemble size	Specify a list of integers. The ensemble GBT algorithm applies the concept of bootstrap aggregation, or "bagging," to GBTs. The training table is sampled a number of times, and a new GBT is built for each sample. This option specifies the number of GBTs to build.

Table 17: Reproducibility seeds - GBTR

Parameter	Description
Table sampling seed	Specify a long value for the data sampling random number seed. If omitted, a time-based seed will be used.
Dimension sampling Seed	Specify a long value for the dimension sampling random number seed. If omitted, a time-based seed will be used.
Model validation sampling seed	Optionally specify the model validation sampling seed. This serves as the holdout sampling random number seed and/or the seed for the number of folds. If omitted, a time-based seed will be used
Smart search sampling seed	Optionally specified a seed value for smart search. If omitted, a time-based seed will be used by default.

Table 18: PMML output - GBTR

Parameter	Description
Enable PMML	Optionally select this checkbox if you want to produce PMML (Predictive Model Markup Language) output.

GLM Options

The following options are available when building GLM models.

Table 19: Smart search - GLM

Parameter	Description
Smart search	Specify to use an intelligent tuning technique (instead of naive grid search). If specified, then only model validation options must be specified (Number of folds and/or Holdout ratio). Users have the option to specify additional tuning parameters, but this is not required. Skytree will determine the best parameter settings to use during tuning. If this is specified, then the Search iterations option becomes available.
Search iterations	If Smart search is enabled, then optionally specify the number of search rounds to try for tuning. This value must be greater than 0 and defaults to 100.

Table 20: Model validation type options - GLM

Parameter	Description
Number of folds (k)	Optionally specify the number of folds for k-fold cross-validation.
Holdout ratio	Specify the fraction of the table to be held out for tuning. This value defaults to 0.2.
Monte Carlo cross-validation	Monte Carlo cross-validation can be used specifying a Holdout ratio in conjunction with the number of folds. With Monte Carlo cross validation, for example, if the number of folds is 10 and the holdout ratio is 0.2, then a random 20% of the training data would be heldout in each fold as the validation set.

Table 21: Method parameters - GLM. Tunable options can be specified as a single value, a comma separated list, or as an array

Parameter	Description
Link function	Specify the link function to use for the GLM model. Options include: • Logit • Complementary log-log
L1 penalty	Specify the penalty value for L1 (LASSO) regularization. This value must be ≥ 0.0 . Specifying both L1 and L2 values > 0 will result in Elastic Net regularization.
L2 penalty	Specify the penalty value for L2 (Ridge) regularization. This value must be ≥ 0.0 . Specifying both L1 and L2 values > 0 will result in Elastic Net regularization.
Tolerance (epsilon)	Specify the numerical accuracy to which to train the model.

Table 22: Advanced method parameters - GLM

Exclude bias term	By default, a bias or intercept term is included in the computed model. The bias term adjusts the log-odds up or down by a constant amount, i.e. it is the predicted log-odds when all inputs are exactly 0. The bias term can be excluded (forced to equal 0) with this option.

Table 22: Advanced method parameters - GLM (continued)

Maximum number of iterations	Specify the maximum allowed number of iterations for the training algorithm. This value must be > 0.
Maximum cache memory in MB	Specify the maximum allowed memory, in megabytes, used to cache the rows of the kernel matrix (to reduce recomputation).

Table 23: Reproducibility seeds - GLM

Parameter	Description
Table sampling seed	Specify a long value for the data sampling random number seed. If omitted, a time-based seed will be used.
Model validation sampling seed	Optionally specify the model validation sampling seed. This serves as the holdout sampling random number seed and/or the seed for the number of folds. If omitted, a time-based seed will be used
Smart search sampling seed	Optionally specified a seed value for smart search. If omitted, a time-based seed will be used by default.

Table 24: PMML output - GLM

Parameter	Description
Enable PMML	Optionally select this checkbox if you want to produce PMML (Predict Model Markup Language) output.

GLMR Options

The following options are available when building GLMR models.

Table 25: Smart search - GLMR

Parameter	Description
Smart search	Specify to use an intelligent tuning technique (instead of naive grid search). If specified, then only model validation options must be specified (Number of folds and/or Holdout ratio). Users have the option to specify additional tuning parameters, but this is not required. Skytree will determine the best parameter settings to use during tuning. If this is specified, then the Search iterations option becomes available.
Search iterations	If Smart search is enabled, then optionally specify the number of search rounds to try for tuning. This value must be greater than 0 and defaults to 100.

Table 26: Model validation type options - GLMR

Parameter	Description
Number of folds (k)	Optionally specify the number of folds for k-fold cross-validation.
Holdout ratio	Specify the fraction of the table to be held out for tuning. This value defaults to 0.2.
Monte Carlo cross-validation	Monte Carlo cross-validation can be used specifying a Holdout ratio in conjunction with the number of folds. With Monte Carlo cross validation, for example, if the number of folds is 10 and the holdout ratio is 0.2, then a random 20% of the training data would be heldout in each fold as the validation set.

Table 27: Method parameters - GLMR. Tunable options can be specified as a single value, a comma separated list, or as an array

Parameter	Description
Family function	Specify the family function to use for a GLMR model. Options include: • Gamma • Gaussian • Poisson • Tweedie
Link function	Specify the link function to use for the GLMR model. Options include: • Canonical. This can be specified for any Family Function. • Identity. This can be specified if the Family Function is Gaussian or Poisson. • Inverse. This can be specified if the Family Function is Gamma. • Log. This can be specified if the Family Function is Gamma, Poisson, or Tweedie. • Square root. This can be specified if the Family Function is Poisson.
L1 penalty	Specify the penalty value for L1 (LASSO) regularization. This value must be ≥ 0.0 . Specifying both L1 and L2 values > 0 will result in Elastic Net regularization.
L2 penalty	Specify the penalty value for L2 (Ridge) regularization. This value must be ≥ 0.0 . Specifying both L1 and L2 values > 0 will result in Elastic Net regularization.
Tolerance (epsilon)	Specify the numerical accuracy to which to train the model.
Tweedie exponent	When a Tweedie family function is specified, then you must also specify a value for the exponent. This value must be > 1.0 and < 2.0 .

Table 28: Advanced method parameters - GLMR

Exclude bias term	By default, a bias or intercept term is included in the computed model. The bias term adjusts the log-odds up or down by a constant amount, i.e. it is the predicted log-odds when all inputs are exactly 0. The bias term can be excluded (forced to equal 0) with this option. Note: This option cannot be used with the following family-link pairs: Gamma-Inverse, Gamma-Canonical, Poisson-Identity, or Poisson-Sqrt.
Maximum number of iterations	Specify the maximum allowed number of iterations for the training algorithm. This value must be > 0.
Maximum cache memory in MB	Specify the maximum allowed memory, in megabytes, used to cache the rows of the kernel matrix (to reduce recomputation).

Table 29: Reproducibility seeds - GLMR

Parameter	Description
Table sampling seed	Specify a long value for the data sampling random number seed. If omitted, a time-based seed will be used.
Model validation sampling seed	Optionally specify the model validation sampling seed. This serves as the holdout sampling random number seed and/or the seed for the number of folds. If omitted, a time-based seed will be used.
Smart search sampling seed	Optionally specified a seed value for smart search. If omitted, a time-based seed will be used by default.

Table 30: PMML output - GLMR

Parameter	Description
Enable PMML	Optionally select this checkbox if you want to produce PMML (Predict Model Markup Language) output.

RDF Options

The following options are available when building RDF models.

Table 31: Model validation type options - RDF

Parameter	Description
Number of folds (k)	Optionally specify the number of folds for k-fold cross-validation.
Holdout ratio	Specify the fraction of the table to be held out for tuning. This value defaults to 0.2.

Table 31: Model validation type options - RDF (continued)

Parameter	Description
Monte Carlo cross-validation	Monte Carlo cross-validation can be used specifying a Holdout ratio in conjunction with the number of folds. With Monte Carlo cross validation, for example, if the number of folds is 10 and the holdout ratio is 0.2, then a random 20% of the training data would be heldout in each fold as the validation set.

Table 32: Method parameters - RDF. Tunable options can be specified as a single value, a comma separated list, or as an array.

Parameter	Description
Number of trees	Specify a list of numbers to indicate the number of trees in the ensemble. This value must be > 0.
Number of dimensions	Specify a list of integers for the number of dimensions to sample at each node. If omitted, $(\log_2(D) + 1)$ is used, where D is the number of attributes.
Probability aggregation method	Specify the method used for aggregation of individual tree predictions. Options include <code>Vote</code> and <code>Average</code> (default).
Sampling ratio	Specify a list of doubles to indicate the per-class sampling ratio. You can use a single <code>Sampling Ratio</code> that will be used for each class, or specify <code>Sampling Ratio</code> values for each class. If a value is not specified, then full bootstrapping will be used.
Sample with replacement	Specifies that for each tree, sample the training points using replacement (bootstrap). This option is on by default. If turned off, then a <code>Sampling ratio</code> value must be provided.
Imbalance	When enabled, Skytree will attempt to improve classification for imbalanced classes. Note that this option cannot be used with <code>Sampling ratio</code> .
Imbalance scale	Specify the relative size of all classes with respect to the smallest class. Requires <code>Imbalance</code> . If <code>Imbalance</code> is specified, then this value defaults to 1.0.

Table 33: Decision tree parameters - RDF. Tunable options can be specified as a single value, a comma separated list, or as an array

Parameter	Description
Impurity	Specify the type of impurity used to generate the split. Specify <code>Entropy</code> (default), <code>Gini</code> , or <code>Misclassification</code> .
Tree depth	Specify the depth to which each ensemble is built. If set to 0, trees will be built to the fullest extent.
Split criterion	Specify whether to use information gain ratio vs. information gain as a measure for split criterion.
Smoothing	Specify the probability smoothing parameter. This value must be ≥ 0 . Note that specifying 0 disables this feature.

Table 33: Decision tree parameters - RDF. Tunable options can be specified as a single value, a comma separated list, or as an array (continued)

Parameter	Description
Categorical selection method	Specify one of the following methods. ▪ Random - Create randomly distributed, equally sized splits on the attribute values ▪ Random exact (default) - For unique attribute values less than or equal to the Categorical random split threshold, this performs exact search. For a higher number of values, random splitting is attempted. ▪ One vs. all - Splits are created with single attribute value on one side and all other values on the other side. For example: blue or not blue ▪ One vs. all random - Tries both Random and One vs. all ▪ One vs. all exact - For unique attribute values less than or equal to the Categorical random split threshold, this performs an exact search. For a higher number of values, One vs. all splitting is attempted. ▪ Exact - Search for the best partition possible. This can be very expensive for categorical attributes with a large number of unique attribute values (except for the case of RDF binary classification). If specified for data with 10 or less unique attribute values, the exhaustive search is performed. For a larger number of values, a heuristic search is performed.
Categorical random split tries	Specify an integer for the number of times to try random categorical splits. This value defaults to 10.
Categorical random split threshold	If the number of categorical values exceeds this integer value, then random splitting is attempted. Larger values will slow down training time considerably for exhaustive searches of all combinations of $O(2^{q-1})$. For binary classification problems, though, it is $O(q)$ and, therefore, can be set to larger values. This value defaults to 4.

Table 34: Advanced method parameters - RDF

Parameter	Description
Sample categorical dimensions using cardinality	Categorical dimensions with larger number of unique values in a given node will have a proportionately higher chance of being selected when sampling dimensions. Continuous features are treated as having one unique value. When off (default), all dimensions have an equal chance of being selected in dimension sampling

Table 35: Reproducibility seeds - RDF

Parameter	Description
Table sampling seed	Specify a long value for the data sampling random number seed. If omitted, a time-based seed will be used.
Dimension sampling seed	Specify a long value for the dimension sampling random number seed. If omitted, a time-based seed will be used.

Table 35: Reproducibility seeds - RDF (continued)

Parameter	Description
Model validation sampling seed	Optionally specify the model validation sampling seed. This serves as the holdout sampling random number seed and/or the seed for the number of folds. If omitted, a time-based seed will be used
Categorical sampling seed	Specify a long value for the categorical values sampling random number seed. If omitted, a time-based seed will be used.

Table 36: PMML output - RDF

Parameter	Description
Enable PMML	Optionally select this checkbox if you want to produce PMML (Predictive Model Markup Language) output.

RDFR Options

The following options are available when building RDFR models.

Table 37: Model validation type options - RDFR

Parameter	Description
Number of folds (k)	Optionally specify the number of folds for k-fold cross-validation.
Holdout ratio	Specify the fraction of the table to be held out for tuning. This value defaults to 0.2.
Monte Carlo cross-validation	Monte Carlo cross-validation can be used specifying a Holdout ratio in conjunction with the number of folds. With Monte Carlo cross validation, for example, if the number of folds is 10 and the holdout ratio is 0.2, then a random 20% of the training data would be heldout in each fold as the validation set.

Table 38: Method parameters - RDFR. Tunable options can be specified as a single value, a comma separated list, or as an array.

Parameter	Description
Number of trees	Specify a list of numbers to indicate the number of trees in the ensemble. This value must be > 0.
Number of dimensions	Specify a list of integers for the number of dimensions to sample at each node. If omitted, $(\log_2(D) + 1)$ is used, where D is the number of attributes.
Sampling ratio	Specify a list of doubles to indicate the per-class sampling ratio. You can use a single Sampling Ratio that will be used for each class, or specify Sampling Ratio values for each class. If a value is not specified, then full bootstrapping will be used.
Sample with replacement	Specifies that for each tree, sample the training points using replacement (bootstrap). This option is on by default. If turned off, then a Sampling ratio value must be provided.

Table 39: Decision tree parameters - RDFR. Tunable options can be specified as a single value, a comma separated list, or as an array.

Tree depth	Specify the depth to which each ensemble is built. If set to 0, trees will be built to the fullest extent.
Minimum node weight	Specify the minimum bound on the total weight in each leaf during tree building.

Table 40: Advanced method parameters - RDFR

Parameter	Description
Sample categorical dimensions using cardinality	Categorical dimensions with larger number of unique values in a given node will have a proportionately higher chance of being selected when sampling dimensions. Continuous features are treated as having one unique value. When off (default), all dimensions have an equal chance of being selected in dimension sampling

Table 41: Reproducibility seeds - RDFR

Parameter	Description
Table sampling seed	Specify a long value for the data sampling random number seed. If omitted, a time-based seed will be used.
Dimension sampling seed	Specify a long value for the dimension sampling random number seed. If omitted, a time-based seed will be used.
Model validation sampling seed	Optionally specify the model validation sampling seed. This serves as the holdout sampling random number seed and/or the seed for the number of folds. If omitted, a time-based seed will be used

Table 42: PMML output - RDFR

Parameter	Description
Enable PMML	Optionally select this checkbox if you want to produce PMML (Predictive Model Markup Language) output.

SVM Options

The following options are available when building SVM models.

Table 43: Smart search - SVM

Parameter	Description
Smart search	Specify to use an intelligent tuning technique (instead of naive grid search). If specified, then only model validation options must be specified (Number of folds and/or Holdout ratio). Users have the option to specify additional tuning parameters, but this is not required. Skytree will determine the best parameter settings to use during tuning. If this is specified, then the Search iterations option becomes available. Note: An SVM smart search run is based on a specified kernel type. If a Kernel type is not specified, smart search will run using a kernel type of Linear. If a different kernel function is desired in smart search, then it must be explicitly specified.
Search iterations	If Smart search is enabled, then optionally specify the number of search rounds to try for tuning. This value must be greater than 0 and defaults to 100.

Table 44: Model validation type options - SVM

Parameter	Description
Number of folds (k)	Optionally specify the number of folds for k-fold cross-validation.
Holdout ratio	Optionally specify the fraction of the table to be held out for tuning.
Monte Carlo cross-validation	Monte Carlo cross-validation can be used specifying a Holdout ratio in conjunction with the number of folds. With Monte Carlo cross validation, for example, if the number of folds is 10 and the holdout ratio is 0.2, then a random 20% of the training data would be heldout in each fold as the validation set.

Table 45: Method parameters - SVM. Tunable options can be specified as a single value, a comma separated list, or as an array

Parameter	Description
Kernel	Specify the kernel function to be used for during SVM tuning and training. Available kernel functions include the following: • Linear (default) • Radial basis function (Gaussian) • Polynomial
Regularization (lambda)	Skytree logistic regression is L1-regularized. The default value for the regularization term is set to 1e - 6. This value is sufficient to completely fit the model to the data. Higher values can be used to ensure under-fit models that may suit the problem better. Altering the regularization term will penalize the L1-norm of the coefficients in the logistic regression. This option is required when training or tuning without smart search. If smart search is enabled, then this value is optional.
Tolerance (epsilon)	Specify the numerical accuracy to which to train the SVM model.
RBF bandwidth	If the Kernel type is specified as Radial basis function (Gaussian) , then optionally specify the bandwidth of the RBF kernel. This value must be >= 0.

Table 45: Method parameters - SVM. Tunable options can be specified as a single value, a comma separated list, or as an array (continued)

Parameter	Description
Polynomial degree	If the Kernel type is specified as Polynomial, then specify the degree of the polynomial kernel. This value must be an integer > 1.
Polynomial scale	If the Kernel type is specified as Polynomial, then specify the scale of the polynomial kernel. This value must be > 0.
Polynomial offset	If the Kernel type is specified as Polynomial, then specify the offset of the polynomial kernel. This value must be >= 0.

Table 46: Advanced method parameters - SVM

Parameter	Description
Exclude bias term	By default, a bias or intercept term is included in the computed model. The bias term adjusts the log-odds up or down by a constant amount, i.e. it is the predicted log-odds when all inputs are exactly 0. The bias term can be excluded (forced to equal 0) with this option.
Maximum number of iterations	Specify the maximum allowed number of iterations for the training algorithm. This value must be > 0.
Maximum cache memory in MB	Specify the maximum allowed memory, in megabytes, used to cache the rows of the kernel matrix (to reduce recomputation).

Table 47: Reproducibility seeds - SVM

Parameter	Description
Table sampling seed	Specify a long value for the data sampling random number seed. If omitted, a time-based seed will be used.
Dimension sampling Seed	Specify a long value for the dimension sampling random number seed. If omitted, a time-based seed will be used.
Model validation sampling seed	Optionally specify the model validation sampling seed. This serves as the holdout sampling random number seed and/or the seed for the number of folds. If omitted, a time-based seed will be used
Smart search sampling seed	Optionally specified a seed value for smart search. If omitted, a time-based seed will be used by default.

Table 48: PMML output - SVM

Parameter	Description
Enable PMML	Optionally select this checkbox if you want to produce PMML (Predict Model Markup Language) output.

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